Probability Notes

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1 Events and their Probabilities

Definition 1.1 (Sample Space)**.** The set of all possible outcomes of an experiment is called the **sample space** and is denoted by Ω. An **event** is a subset of the sample space.

Example 1.1. Consider an experiment of flipping of a coin three times. The sample space is

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Let us define an event A indicating that majority of coins shows up head. Therefore

$$A = \{HHH, HHT, HTH, THH\}.$$

Example 1.2. Consider an experiment of rolling two dice. The sample space is

$$\begin{aligned} S = \{&(1,1),(1,2),\ldots,(1,6),\\ &(2,1),(2,2),\ldots,(2,6),\\ &\vdots \quad \vdots \quad \ldots \quad \vdots\\ &(6,1),(6,2),\ldots,(6,6)\}. \end{aligned}$$

Let us define an event A indicating that sum of the faces showing is 7. Therefore

$$A = \{(1,6),(2,5),(5,2),(3,4),(4,3),(6,1)\}.$$

There are three ways to indicate an experimenter's sample space:

1. If the number of possible outcomes is small then listing can be done.
2. In some cases structure can be shown.
3. State a mathematical formula that sample outcomes must satisfy.
   For example a programmer is running a subroutine that solves a general quadratic equation $ax^2 + bx + c = 0$. Experiment consist of choosing the values for three coefficients a, b and c. In this case $S = \{(a,b,c) : \infty < a < \infty, \infty < b < \infty, \infty < c < \infty,\}$. If the event A: Equation has two equal roots, then $A = \{(a,b,c) : b^2 - 4ac = 0\}$.

There are three very different ways of defining probability of an event.

1. **Classical Definition** : Imagine an experiment, or game, having n possible outcomes-and suppose that those outcomes are equally likely. If some event A were satisfied by m of those n then

   $$P(A) = \frac{m}{n}.$$

   Limitations of this definition are
   (i) What if outcomes of the experiments are not equally likely? or
   (ii) the number of outcomes is not finite.

2. **Empirical Definition**: Consider a sample space, S, and any event, A, defined on S. If experiment were performed one time, either A or A^c would be the outcome. If it were performed n times, and let us suppose that m times A has occurred $0 \leq m \leq n$. Now repeat the process an infinite number of times. As n gets large, the ratio m/n will fluctuate less and less, and it will coverage to some number. This number is called $P(A)$, that is,

$$P(A) = \lim_{n\to\infty} \frac{m}{n}.$$

It is shown in the following figure:
For example, if a fair coin were to be tossed repeatedly with A being the event "Head appears". It is reasonable to expect that $P(A) = \lim_{n\to\infty} \frac{m}{n}$ would be equal to 1/2.

There are some limitations of this definition e.g., (i) how large n must be to give a good approximation for $P(A)$? (ii) what if the coin is biased? (iii) repeating the experiment under identical conditions for an infinite number of times is physically impossible.

Before we discuss next definition, it is observed that, probability is not defined as a ratio, or as a limit to the ratio, but simply as a function that behaved in accordance with a prescribed set of axioms.

3. **Axiomatic Definition**: If S has a finite number of members, the necessary and sufficient set of axioms for P are:
Axiom 1: Let A be any event defined over S. Then $P(A) \geq 0$.
Axiom 2: $P(S) = 1$
Axiom 3: If A and B be any two mutually exclusive (disjoint) events defined over S. Then

$$P(A \cup B) = P(A) + P(B). \tag{1}$$

When S has an infinite number of members, a fourth axioms is needed:
Axiom 4: Let $A_1, A_2, \ldots,$ be events defined over S. If $A_i \cap B_j = \Phi$ for each $i \neq j$, then

$$P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i). \tag{2}$$

The axiomatic definition of probability can be described/generalised from the perspective of measure as below.

Definition 1.2 (Field)**.** Any collection $\mathscr{F}$ of subsets of Ω which satisfies the following three conditions is a **field**:

1. if $A, B \in \mathscr{F}$ then $A \cup B \in \mathscr{F}$ and $A \cap B \in \mathscr{F}$,
2. if $A \in \mathscr{F}$ then $A^c \in \mathscr{F}$,
3. the empty set ϕ belongs to $\mathscr{F}$.

Definition 1.3 ($\sigma-$ field)**.** A collection $\mathscr{F}$ of subsets of Ω is called a $\sigma-$ field if it satisfies the following:

1. $\phi \in \mathscr{F}$,
2. if $A_1, A_2, \ldots \in \mathscr{F}$ then $\cup_{i=1}^{\infty} \in \mathscr{F}$,
3. if $A \in \mathscr{F}$ then $A^c \in \mathscr{F}$.

Definition 1.4 (Probability Space)**.** A **probability measure** on $(\Omega, \mathscr{F})$ is a function $\mathbb{P} : \mathscr{F} \to [0, 1]$ satisfying

1. $\mathbb{P}(\phi) = 0, \mathbb{P}(\Omega) = 1$,
2. if $A_1, A_2, \ldots$ is a collection of disjoint events of $\mathscr{F}$ such that $A_i \cap A_j = \phi \ \forall i \neq j, then$

$$\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

$(\Omega, \mathscr{F}, \mathbb{P})$ is called the **probability space**.

Theorem 1.1. If A and B are two disjoint events in Ω, then

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$,
2. if $B \supseteq A$, then $\mathbb{P}(B) \geq \mathbb{P}(A)$,
3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$,

Proof: (1) From definition 1.4, $\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(A \cup A^c) = \mathbb{P}(\Omega) = 1$, Thus, $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$.
(2) $\mathbb{P}(B) = \mathbb{P}(A \cup (B \backslash A)) = \mathbb{P}(A) + \mathbb{P}(B \backslash A) \geq \mathbb{P}(A)$, since $\mathbb{P}(B \backslash A) \geq 0$.
(3) We know that A can be written as a disjoint union of $A \backslash B, B \backslash A$, and $A \cup B$, then

$$\begin{aligned}
\mathbb{P}(A \cup B) &= \mathbb{P}(A \backslash B) + \mathbb{P}(B \backslash A) + \mathbb{P}(A \cap B) \\
&= \mathbb{P}(A \backslash (A \cap B)) + \mathbb{P}(B \backslash (A \cup B)) + \mathbb{P}(A \cap B) \\
&= \mathbb{P}(A) - \mathbb{P}(A \cup B) + \mathbb{P}(B) - \mathbb{P}(A \cup B) + \mathbb{P}(A \cup B) \\
&= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).
\end{aligned}$$

Theorem 1.2 (**Principle of Inclusion and Exclusion**)**.** For events $A_1, A_2, \ldots, A_n$ in a probability space $(\Omega, \mathscr{F}, \mathbb{P})$,

$$\begin{aligned}
\mathbb{P}(\cup_{i=1}^{n})A_i = \sum_{i=1}^{n} \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} \mathbb{P}(A_i \cap A_j \cap A_k) + \cdots \\
+ (-1)^{n-1} \sum_{1 \leq i < \ldots j \leq n} \mathbb{P}(\cap_{i=1}^{n} A_i).
\end{aligned}$$

Proof: Proof by induction on n.

Theorem 1.3 (**Boole's Inequality**). For events $A_1, A_2, \ldots, A_n$ in a probability space $(\Omega, \mathscr{F}, \mathbb{P})$,

$$\mathbb{P}(\cup_{i=1}^n A_i) \leq \sum_{i=1}^n \mathbb{P}(A_i), \quad \mathbb{P}(\cap_{i=1}^n A_i) \geq 1 - \sum_{i=1}^n \mathbb{P}(A_i^c).$$

Proof: Proof by induction on n.

Definition 1.5 (Independent Events). Events A and B are called **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

More generally, a family of events $\{A_i : i \in I\}$ are independent if

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$$

for all finite subsets J of I.

Theorem 1.4. If A and B are independent, then the following pair of events are also independent:

(a) A, B^c

(b) A^c, B

(c) A^c, B^c

Proof: A and B are independent implies $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Thus,

$$\begin{aligned}
(a)\ \mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A)[1 - \mathbb{P}(B)] = \mathbb{P}(A)\mathbb{P}(B^c), \\
(b)\ \mathbb{P}(A^c \cap B) &= \mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(B)[1 - \mathbb{P}(A)] = \mathbb{P}(A^c)\mathbb{P}(B), \\
(c)\ \mathbb{P}(A^c \cap B^c) &= \mathbb{P}((A \cup B)^c) = 1 - \mathbb{P}(A \cup B) = 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A \cap B) \\
&= \mathbb{P}(A^c) - \mathbb{P}(B) + \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A^c) - \mathbb{P}(B)\mathbb{P}(A^c) \\
&= \mathbb{P}(A^c)[1 - \mathbb{P}(B)] = \mathbb{P}(A^c)\mathbb{P}(B^c).
\end{aligned}$$

Theorem 1.5. (a)Let $A_1, A_2, \ldots$ be an increasing sequence of events, so that $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$, and write A for their limit:

$$A = \bigcup_{i=1}^{\infty} A_i = \lim_{i \to \infty} A_i.$$

Then $\mathbb{P}(A) = \lim_{i \to \infty} \mathbb{P}(A_i)$.
(b) Let $B_1, B_2, \ldots$ be a decreasing sequence of events, so that $B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots$, and write B for their limit:

$$B = \bigcap_{i=1}^{\infty} B_i = \lim_{i \to \infty} B_i.$$

Then $\mathbb{P}(B) = \lim_{i \to \infty} \mathbb{P}(B_i)$.

Proof: (a) We can write $A = A_1 \cup A_2 \cup A_3 \cup \ldots = A_1 \cup (A_2 \backslash A_1) \cup (A_3 \backslash A2) \cap \ldots$. Thus,

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A_1) + \mathbb{P}(A_2 \backslash A_1) + \mathbb{P}(A_3 \backslash A_2) + \ldots \\ &= \mathbb{P}(A_1) + \mathbb{P}(A_2) - \mathbb{P}(A_1) + \mathbb{P}(A_3) - \mathbb{P}(A_2) + \ldots \\ &= \lim_{i\to\infty} \mathbb{P}(A_i).\end{aligned}$$

(b) Since $B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots$, then $B_1^c \subseteq B_2^c \subseteq B_3^c \subseteq \cdots$. Thus, by part (1) of this theorem

$$\mathbb{P}(B) = 1 - \mathbb{P}(B^c) = 1 - \lim_{i\to\infty} B_i^c = 1 - \lim_{i\to\infty}(1 - B_i) = \lim_{i\to\infty} B_i.$$

First- Borel Cantelli Lemma: If $(A_n)_n$ is a sequence of events such that $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(limsup_n A_n) = 0$.
Proof: Let $B_m = \cup_{n\geq m} A_n$ then $B_{m+1} \subset B_m \ \forall\ m$, and hence

$$\begin{aligned}\mathbb{P}(limsup_n A_n) &= \mathbb{P}(\cap_{m\geq 1}(\cup_{n\geq m} A_n)) = \mathbb{P}(\cap_{m\geq 1} B_m) = \lim_{m\to\infty} \mathbb{P}(B_m) \\ &= \lim_{m\to\infty} \mathbb{P}(\cup_{n\geq m} A_n) \leq \lim_{m\to\infty}(\mathbb{P}(A_m) + \mathbb{P}(A_{m+1}) + \ldots) \\ &= \lim_{m\to\infty}\left(\sum_{n=1}^{\infty}\mathbb{P}(A_n) - \sum_{n=1}^{m-1}\mathbb{P}(A_n)\right) = \sum_{n=1}^{\infty}\mathbb{P}(A_n) - \lim_{m\to\infty}\left(\sum_{n=1}^{m-1}\mathbb{P}(A_n)\right) \\ &= \sum_{n=1}^{\infty}\mathbb{P}(A_n) - \sum_{n=1}^{\infty}\mathbb{P}(A_n) = 0.\end{aligned}$$

The second last statement is justified by the fact that $\sum_{n=1}^{\infty} \mathbb{P}(A_n)$ converges.

Second- Borel Cantelli Lemma: If the sequence of events $(A_n)_n$ is independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(limsup_n(A_n)) = 1$.
Proof: Since $\mathbb{P}(A_n) \geq 0$ implies $1 - \mathbb{P}(A_n) \leq e^{-\mathbb{P}(A_n)}$, for $k \leq m$,

$$\mathbb{P}(\cap_{n=k}^{m} A_n^c) = \prod_{n=k}^{m}\mathbb{P}(A_n^c) = \prod_{n=k}^{m}(1 - \mathbb{P}(A_n)) \leq \prod_{n=k}^{m} e^{-\mathbb{P}(A_n)} = e^{-\sum_{n=k}^{m}\mathbb{P}(A_n)},$$

$$\begin{aligned}1 - \mathbb{P}(\cup_{n=k}^{\infty} A_n) &= 1 - \lim_{m\to\infty}\mathbb{P}(\cup_{n=k}^{m} A_n) = \lim_{m\to\infty}(1 - \mathbb{P}(\cup_{n=k}^{m} A_n)) \\ &= \lim_{m\to\infty}\mathbb{P}((\cup_{n=k}^{m} A_n)^c) \lim_{m\to\infty}\mathbb{P}(\cap_{n=k}^{\infty} A_n^c) \\ &\leq \lim_{m\to\infty} e^{-\sum_{n=k}^{m}\mathbb{P}(A_n)} = e^{-\lim_{m\to\infty}\sum_{n=k}^{m}\mathbb{P}(A_n)} \\ &= e^{-\infty} = 0.\end{aligned}$$

Thus, $\mathbb{P}(\cup_{n=k}^{\infty} A_n) = 1 \ \forall\ k$, and hence

$$1 = \lim_{k\to\infty}\mathbb{P}(\cup_{n=k}^{\infty} A_n) = \mathbb{P}(\cap_{k=1}^{\infty}(\cup_{n=k}^{\infty} A_n)) = \mathbb{P}(limsup_n(A_n)).$$

Definition 1.6 (Conditional Probability)**.** If $\mathbb{P}(B) > 0$ then the **conditional probability** that A occurs given that B occurs is defined to be

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Theorem 1.6. (a) For any events A and B such that $0 < \mathbb{P}(B) < 1$,

$$\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c).$$

(b) More generally, let $B_1, B_2, \ldots, B_n$ be a partition of A such that $\mathbb{P}(B_i) > 0$ for all i. Then

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

Proof: (a) We can write A as a disjoint union of $A \cap B$ and $A \cap B^c$. Then using the definition of conditional probability

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \\ &= \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c).\end{aligned}$$

(b) We can write A as a disjoint union of $A \cap B_i$ where $i \in \{1, 2, \ldots, n\}$. Then using the definition of conditional probability

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap B_1) + \mathbb{P}(A \cap B_1) + \ldots + \mathbb{P}(A \cap B_n) \\ &= \mathbb{P}(A|B_1)\mathbb{P}(B_1) + \mathbb{P}(A|B_2)\mathbb{P}(B_2) + \ldots + \mathbb{P}(A|B_n)\mathbb{P}(B_n).\end{aligned}$$

Theorem 1.7 (**Bayes Theorem**)**.** Let $A_1, A_2, \ldots, A_n$ be a partition of Ω each A_i having positive probability, then for any event B,

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}.$$

Proof: Using the definition of conditional probability and last theorem, we can write

$$\begin{aligned}\mathbb{P}(A_j|B) &= \frac{\mathbb{P}(A_j \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A_j \cap B)}{\mathbb{P}((B \cap A_1) \cup (B \cap A_2) \cup \cdots \cup (B \cap A_n))} \\ &= \frac{\mathbb{P}(A_j \cap B)}{\sum_{i=1}^{n} \mathbb{P}(B \cap A_i)} = \frac{\mathbb{P}(B|A_j)P(A_j)}{\sum_{i=1}^{n} \mathbb{P}(B|A_j)P(A_j)}.\end{aligned}$$

Example 1.3. A rare disease affects one person in 10^5. A test for the disease shows $+ve$ with probability $99/100$ when applied to an ill person, and with probability $1/100$ when applied to an healthy person. What is the probability that you have the disease given that the test shows $+ve$?

$$\begin{aligned}\mathbb{P}(ill| + ve) &= \frac{\mathbb{P}(+ve|ill)\mathbb{P}(ill)}{\mathbb{P}(+ve|ill)\mathbb{P}(ill) + \mathbb{P}(+ve|healthy)\mathbb{P}(healthy)} \\ &= \frac{\frac{99}{100} \times \frac{1}{10^5}}{\frac{99}{100} \times \frac{1}{10^5} + \frac{1}{100} \times \frac{10^5-1}{10^5}} = \frac{99}{100098}.\end{aligned}$$

2 Random Variables and their Distribution

Definition 2.1 (Random Variable)**.** A **random variable** is a function $X : \Omega \to \mathbb{R}$ with the property that $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathscr{F}$ for each $x \in \mathbb{R}$. Such a function is said to be $\mathscr{F}-$ measurable.

Definition 2.2 (Distribution Function)**.** The **distribution function** of a random variable X is the function $F : \mathbb{R} \to [0, 1]$ given by

$$F_X(x) = \mathbb{P}(X \leq x); \forall\ x \in \mathbb{R}.$$

Example 2.1. Let $\Omega = HH, HT, TH, TT$ be a sample space of tossing a coin twice where H and T represents head and tail, respectively. Then $X : \Omega \to \mathbb{R}$ defined by $X(\omega) = no.\ of\ heads\ in\ \omega$, is a random variable where

$$X = \begin{cases} 0 & with\ probability\ 1/4 \\ 1 & with\ probability\ 1/2 \\ 2 & with\ probability\ 1/4. \end{cases}$$

The distribution function of the random variable X is given by

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1/4 & 0 \leq x < 1 \\ 3/4 & 1 \leq x < 2 \\ 1 & x > 2. \end{cases}$$

Theorem 2.1. Let F be the distribution function of a random variable X, then

(a) $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$,

(b) if $x < y$, then $F(x) \leq F(y)$,

(c) F is right-continuous function, that is, $F(x + h) \to F(x)$ as $h \downarrow 0$,

Proof: (a) Let $B_n = \{\omega \in \Omega : X(\omega) \leq -n\}$, then $B_1 \supseteq B_2 \supseteq B_3 \supseteq \ldots$ with $\lim_{n\to\infty} B_n = \phi$. Then

$$\lim_{x\to-\infty} F_X(x) = \lim_{x\to-\infty} \mathbb{P}(X \leq x) = \lim_{n\to\infty} \mathbb{P}(B_n) = \mathbb{P}(\phi) = 0.$$

Let $A_n = \{\omega \in \Omega : X(\omega) \leq n\}$, then $A_1 \subseteq A_2 \subseteq A_3 \subseteq \ldots$ with $\lim_{n\to\infty} A_n = \Omega$. Then

$$\lim_{x\to\infty} F_X(x) = \lim_{x\to\infty} \mathbb{P}(X \leq x) = \lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(\Omega) = 1.$$

(b) For $x < y$, let $A_x = \{\omega \in \Omega : X(\omega) \leq x\}, A_y = \{\omega \in \Omega : X(\omega) \leq y\}$, and $A_{xy} = \{\omega \in \Omega : x < X(\omega) \leq y\}$, then $A_y = A_x \cup A_{xy}$. Since A_x and A_{xy} are disjoint, $\mathbb{P}(A_y) = \mathbb{P}(A_x) + \mathbb{P}(A_{xy})$, i.e.,

$$F(y) = F(x) + \mathbb{P}(x < X \leq y) \geq F(x).$$

(c) For $x \in \mathbb{R}$, let $A_n = \{\omega \in \Omega : X(\omega) \leq x + \frac{1}{n}\}$ and $B_n = \{\omega \in \Omega : X(\omega) \leq x - \frac{1}{n}\}$. Since A_n is a decreasing sequence of events with $\lim_{n\to\infty} A_n = \{\omega \in \Omega : X(\omega) \leq x\}$ and B_n is an increasing sequence of events with $\lim_{n\to\infty} B_n = \{\omega \in \Omega : X(\omega) < x\}$,

$$\begin{aligned}\lim_{h\downarrow 0} F(x+h) &= \lim_{h\downarrow 0} \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x+h\}) = \lim_{n\to\infty} \mathbb{P}(A_n) \\ &= \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = F(x),\end{aligned}$$

and

$$\begin{aligned}\lim_{h\downarrow 0} F(x-h) &= \lim_{h\downarrow 0} \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x-h\}) = \lim_{n\to\infty} \mathbb{P}(B_n) \\ &= \mathbb{P}(\{\omega \in \Omega : X(\omega) < x\}) = F(x^-) \quad (\neq F(x)).\end{aligned}$$

Thus, the distribution function F is right-continuous but not left-continuous.
In fact, a function F is the distribution function of some random variable if and only if it satisfies the properties of the above theorem.

Theorem 2.2. Let F be a distribution function of a random variable X, then:

(a) $\mathbb{P}(X > x) = 1 - F(x)$,

(b) $\mathbb{P}(x < X \leq y) = F(y) - F(x)$,

(c) $\mathbb{P}(X = x) = F(x) - \lim_{y\uparrow x} F(y)$.

Proof: (a) $\mathbb{P}(X \leq x) + \mathbb{P}(X > x) = 1$, implies $\mathbb{P}(X > x) = 1 - \mathbb{P}(X \leq x) = 1 - F(x)$.
(b) For $x < y$,

$$\begin{aligned}\mathbb{P}(x < X \leq y) &= \mathbb{P}(\{\omega \in \Omega : x < X(\omega) \leq y\}) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq y\} \cap \{\omega \in \Omega : x < X(\omega)\}) \\ &= P(\{\omega \in \Omega : X(\omega) \leq y\}) + P(\{\omega \in \Omega : x < X(\omega)\}) \\ &\quad - P(\{\omega \in \Omega : X(\omega) \leq y\} \cup \{\omega \in \Omega : x < X(\omega)\}) \\ &= F(y) + 1 - F(x) - P(\{\omega :\in \Omega : X(\omega) \in \mathbb{R}\}) = F(y) + 1 - F(x) - 1 \\ &= F(y) - F(x).\end{aligned}$$

(c) Let $B_n = \{x - 1/n < X \leq x\}$, then B_n is a decreasing sequence with limit $B = \{X = x\}$. Thus,

$$\begin{aligned}\mathbb{P}(X = x) = \mathbb{P}(B) &= \lim_{n\to\infty} \mathbb{P}(B_n) = \lim_{n\to\infty} \mathbb{P}(\{x - 1/n < X \leq x\}) \\ &= \lim_{n\to\infty} \mathbb{P}(\{X \leq x\} \cap \{X \leq x - 1/n\}^c) \\ &= \lim_{n\to\infty} (\mathbb{P}(\{X \leq x\}) + \mathbb{P}(\{X \leq x - 1/n\}^c) - \mathbb{P}(\{X \leq x\} \cup \{X \leq x - 1/n\}^c)) \\ &= \mathbb{P}(X \leq x) + \lim_{n\to\infty} \mathbb{P}(\{X \leq x - 1/n\}^c) - \lim_{n\to\infty} \mathbb{P}(X \in \mathbb{R}) \\ &= F(x) + 1 - \lim_{n\to\infty} \mathbb{P}(\{X \leq x - 1/n\}) - 1 \\ &= F(x) - \lim_{n\to\infty} F(x - 1/n) = F(x) - \lim_{y\uparrow x} F(y).\end{aligned}$$

Example 2.2. A random variable X has distribution function F. Find the distribution function of $Y = aX + b$, where $a, b \in \mathbb{R}$.
For $a > 0$, $F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(aX + b \leq y) = \mathbb{P}(X \leq \frac{y-b}{a}) = F(\frac{y-b}{a})$.
For $a < 0$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) = F(aX + b \leq y) = \mathbb{P}(aX \leq y - b) = \mathbb{P}\left(X \geq \frac{y-b}{a}\right) \\ &= 1 - \mathbb{P}\left(X < \frac{y-b}{a}\right) = 1 - \mathbb{P}\left(X \leq \frac{y-b}{a}\right) + \mathbb{P}\left(X = \frac{y-b}{a}\right) \\ &= 1 - F\left(\frac{y-b}{a}\right) + \mathbb{P}\left(X = \frac{y-b}{a}\right). \end{aligned}$$

For $a = 0$,

$$F_Y(y) = \begin{cases} 0, & if\ y < b, \\ 1, & if\ y \geq b. \end{cases}$$

Example 2.3. Show that if F and G are distribution functions and $0 \leq \lambda \leq 1$, then $\lambda F + (1 - \lambda)G$ is a distribution.
Let $H(x) = \lambda F(x) + (1 - \lambda)G(x)$.
(a) If $x < y$, then $F(x) \leq F(y)$ and $G(x) \leq G(y)$; this implies $H(x) \leq H(y)$.
(b) $\lim\limits_{x \to -\infty} H(x) = \lambda \lim\limits_{x \to -\infty} F(x) - (1 - \lambda) \lim\limits_{x \to -\infty} G(x) = \lambda \times 0 + (1 - \lambda) \times 0 = 0$,
and $\lim\limits_{x \to \infty} H(x) = \lambda \lim\limits_{x \to \infty} F(x) + (1 - \lambda) \lim\limits_{x \to \infty} G(x) = \lambda \times 1 + (1 - \lambda) \times 1 = 1$.
(c) Since F and G are right continuous, $\lambda F + (1 - \lambda)G$ is right continuous by real analysis.
By theorem 2.2, (a), (b), and (c) proves that $\lambda F + (1 - \lambda)G$ is a distribution function.

Definition 2.3 (Discrete Random Variable)**.** A random variable is called **discrete** if it takes values in some countable subset $\{x_1, x_2, \ldots\}$ of $\mathbb{R}$. The discrete random variable has **probability mass function** $f : \mathbb{R} \to [0, 1]$ given by

$$f(x) = \mathbb{P}(X = x)$$

Definition 2.4 (Continuous Random Variable)**.** The random variable X is called **continuous** if its distribution function can be expressed as

$$F(x) = \int_{-\infty}^{x} f(u)du \ \forall\ x \in \mathbb{R},$$

for some integrable function $f : \mathbb{R} \to [0, \infty)]$ called the **probability density function** of X.

The distribution function of these random variables can be written as:

$$F(x) = \begin{cases} \sum_{i=-\infty}^{x} f(i) & \text{if X is discrete r.v.,} \\ \int_{-\infty}^{x} f(u)du & \text{if X is continuous r.v.} \end{cases}$$

Note that $F(\infty) = 1$, in both of the above cases.

Lemma: The probability mass function $f : \mathbb{R} \to [0, 1]$ satisfies:

(a) the set of x such that $f(x) \neq 0$ is countable,

(b) $\sum_i f(x_i) = 1$ where $x_i's$ are values in $\mathbb{R}$ such that $f(x_i) \neq 0$.

Proof: The proof is obvious.

Definition 2.5 (Random Vector)**.** A **random vector** is n random variables $X_1, X_2, \ldots, X_n$ written in vector form as $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$ which takes values in $\mathbb{R}^n$.

For vectors $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ and $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$, we write $\boldsymbol{x} \leq \boldsymbol{y}$ if $x_i \leq y_i$ for each $i = 1, 2, \ldots, n$.

Definition 2.6 (Joint Distribution Function)**.** The **joint distribution function** of a random vector $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$ on the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ is the function $F_{\boldsymbol{X}} : \mathbb{R}^n \to [0, 1]$ given by $F_{\boldsymbol{X}}(\boldsymbol{x}) = \mathbb{P}(\boldsymbol{X} \leq \boldsymbol{x})$ for $\boldsymbol{x} \in \mathbb{R}^n$.

Theorem 2.3. The joint distribution function of the random vector $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$ has the following properties:

(a) $\lim_{\boldsymbol{x} \to -\infty} F_{\boldsymbol{X}}(\boldsymbol{x}) = 0$, $\lim_{\boldsymbol{x} \to \infty} F_{\boldsymbol{X}}(\boldsymbol{x}) = 1$,

(b) if $\boldsymbol{x} \leq \boldsymbol{y}$, then $F_{\boldsymbol{X}}(\boldsymbol{x}) \leq F_{\boldsymbol{X}}(\boldsymbol{y})$,

(c) $F_{\boldsymbol{X}}$ is continuous from above, in that,

$$F_{\boldsymbol{X}}(\boldsymbol{x} + \boldsymbol{h}) \to F_{\boldsymbol{X}}(\boldsymbol{x}) \text{ as } \boldsymbol{h} \downarrow \boldsymbol{0},$$

where $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$.

Proof: Apply Theorem 2.1 to each coordinate of the random vector to get the result.

Let $F_{X,Y}$ be the joint distribution function of random variables X and Y, then the marginal distribution functions of X and Y, F_X and F_Y, can be obtained from their joint distribution function as follows:

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x \ \& \ -\infty < Y < \infty) = \mathbb{P}(X \leq x \ \& \ Y < \infty) \\ &= F_{X,Y}(x, \infty) = \lim_{y \to \infty} F_{X,Y}(x, y). \end{aligned}$$

Similarly, $F_Y(y) = \lim_{x \to \infty} F_{X,Y}(x, y)$.

3 Discrete Random Variable

Recall that, a random variable is called **discrete** if it takes values in some countable subset $\{x_1, x_2, \ldots\}$ of $\mathbb{R}$. The discrete random variable has **probability mass function** $f : \mathbb{R} \to [0, 1]$ given by

$$f(x) = \mathbb{P}(X = x)$$

Definition 3.1 (Independent Discrete Random Variables)**.** Discrete variables X and Y are **independent** if the events $\{X = x\}$ and $\{Y = y\}$ are independent for all x and y, that is,

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y).$$

Definition 3.2 (Expectation)**.** The **mean value**, or **expectation** of a discrete random variable X with mass function f is defined to be

$$\mathbb{E}(X) = \sum_x x f(x)$$

whenever this sum is absolutely convergent.

Theorem 3.1. The expectation operator $\mathbb{E}$ has the following properties:

(a) if $X \geq 0$ then $\mathbb{E}(X) \geq 0$,

(b) if $a, b \in \mathbb{R}$ then $\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$,

(c) $\mathbb{E}(c) = c$ for a constant $c \in \mathbb{R}$.

Proof: (a) If $X \geq 0$, it takes non-negative values only. Also for X taking any value x, $f(x) = \mathbb{P}(X = x) \geq 0$. Thus, $\mathbb{E}(X) = \sum_x x f(x) \geq 0$.

(b)

$$\begin{aligned}
\mathbb{E}(aX + bY) &= \sum_x \sum_y (ax + by) f(x, y) = \Big(\sum_x ax \sum_y f(x, y)\Big) + \Big(\sum_y by \sum_x f(x, y)\Big) \\
&= \sum_x a x f_X(x) + \sum_y b y f_Y(y) = a \sum_x x f_X(x) + b \sum_y y f_Y(y) \\
&= a\mathbb{E}(X) + b\mathbb{E}(Y).
\end{aligned}$$

(c) $\mathbb{E}(c) = \sum_x x\mathbb{P}(X = c) = c\mathbb{P}(X = c) = c \cdot 1 = c$.

Theorem 3.2. If X and Y are independent, then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.
Proof: Let $f_X, f_Y, f_{X,Y}$ be the mass function of $X, Y, (X, Y)$, respectively. Since X and Y are independent, $f_{X,Y}(x, y) = f_X(x) f_Y(y)$. And hence,

$$\begin{aligned}
\mathbb{E}(XY) &= \sum_x \sum_y xy f_{X,Y}(x, y) = \sum_x \sum_y xy f_X(x) f_Y(y) \\
&= \Big(\sum_x x f_X(x)\Big)\Big(\sum_y y f(y)\Big) = \mathbb{E}(X)\mathbb{E}(Y).
\end{aligned}$$

Definition 3.3 (Uncorrelated Random Variable)**.** X and Y are **uncorrelated** random variables if

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y).$$

Theorem 3.3. If the random variable X has a mass function, and $g : \mathbb{R} \to \mathbb{R}$, then

$$\mathbb{E}(g(X)) = \sum_x g(x)f(x),$$

whenever this sum is absolutely convergent.

Proof: Let $Y = g(X)$, then

$$\begin{aligned}\mathbb{E}[g(X)] = \mathbb{E}[Y] &= \sum_y y f_Y(y) = \sum_y y\mathbb{P}(Y = y) = \sum_y y\mathbb{P}(g(X) = y) \\ &= \sum_y y \sum_{x:g(x)=y} \mathbb{P}(X = x) = \sum_y y \sum_{x:g(x)=y} f_X(x) = \sum_x g(x)f_X(x).\end{aligned}$$

Definition 3.4 (kth moment)**.** If k is a positive integer, the kth **moment** m_k of X is defined to be

$$m_k = \mathbb{E}(X^k).$$

The kth **central moment** σ_k is

$$\sigma_k = \mathbb{E}((X - m_1)^k).$$

Definition 3.5 (Variance)**.** The **variance** of a discrete random variable X is defined to be

$$Var(X) = \mathbb{E}[(X - \mathbb{E}(X))^2].$$

Variance of a random variable X can also be written as:

$$\begin{aligned}Var(X) &= \mathbb{E}[(X - m_1)^2] = \mathbb{E}[X^2 - 2m_1X + m_1^2] = \mathbb{E}(X^2) - 2m_1\mathbb{E}(X) + \mathbb{E}(m_1^2) \\ &= \mathbb{E}(X^2) - 2m_1^2 + m_1^2 = m_2 - m_1^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2.\end{aligned}$$

Theorem 3.4. For random variables X and Y, and $a, b \in \mathbb{R}$,

(a) $Var(aX) = a^2Var(X)$,

(b) $Var(X + Y) = Var(X) + Var(Y)$, if X and Y are uncorrelated.

Proof: (a) Using linearity of $\mathbb{E}$,

$$\begin{aligned}Var(aX) &= \mathbb{E}[(aX - \mathbb{E}[aX])^2] = \mathbb{E}[(aX - a\mathbb{E}[X])^2] = \mathbb{E}[a^2(X - \mathbb{E}[X])^2] \\ &= a^2\mathbb{E}[(X - \mathbb{E}[X])^2] = a^2Var(X).\end{aligned}$$

(b) Since X and Y are uncorrelated, $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$; and hence

$$\begin{aligned}Var(X + Y) &= \mathbb{E}[(X + Y - \mathbb{E}(X + Y))^2] \\ &= \mathbb{E}[(X - \mathbb{E}(X))^2 + (Y - \mathbb{E}(Y))^2 + 2(XY - \mathbb{E}(X)\mathbb{E}(Y))] \\ &= Var(X) + Var(Y) + 2(\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)) \\ &= Var(X) + Var(Y).\end{aligned}$$

Definition 3.6 (Covariance)**.** The **covariance** of two random variables X and Y is defined as:

$$Cov(X,Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))].$$

The covariance can also be wriiten as:

$$\begin{aligned} Cov(X,Y) &= \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] \\ &= \mathbb{E}[XY - \mathbb{E}[X]Y - \mathbb{E}[Y]X + \mathbb{E}[X]\mathbb{E}[Y]] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \end{aligned}$$

If X and Y are independent, $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(\mathbb{Y})$; so $Cov(X,Y) = 0$.

Theorem 3.5. For random variables X and Y, and $a, b \in \mathbb{R}$,

$$Var(aX + bY) = a^2 Var(X) + 2abCov(X,Y) + b^2 Var(Y).$$

Proof: Using the part (a) and (b) of this theorem, we get

$$\begin{aligned} Var(aX + bY) &= \mathbb{E}[(aX + bY - \mathbb{E}[aX + bY])^2] = \mathbb{E}[(aX + bY - a\mathbb{E}[X] - b\mathbb{E}[Y])^2] \\ &= \mathbb{E}[a(X - \mathbb{E}[X]) + b(Y - \mathbb{E}[Y])^2] \\ &= \mathbb{E}[a^2(X - \mathbb{E}[X])^2 + b^2(Y - \mathbb{E}[Y])^2 + 2ab(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= a^2\mathbb{E}[(X - \mathbb{E}[X])^2] + b^2\mathbb{E}[(Y - \mathbb{E}[Y])^2] + 2ab\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= a^2 Var(X) + b^2 Var(Y) + 2abCov(X,Y). \end{aligned}$$

Definition 3.7 (Coorelation)**.** The **correlation coefficient** of random variables X and Y is

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X) \cdot Var(Y)}}$$

as long as the variances are non-zero.

Theorem 3.6. The correlation coefficient ρ satisfies $|\rho(X,Y)| \leq 1$ with equality if and only if $\mathbb{P}(Y = aX + b) = 1$ for some $a, b \in \mathbb{R}$.

Proof: we know that $E\{Y - a\}^2$ is minimized when we choose $a = E(Y)$ so that $E(Y)$ may be interpreted as the best constant predictor of Y. If instead, We choose to predict Y by a linear function of X, say $aX + b$, and measure the error in this prediction by $E\{Y - aX - b\}^2$, then we should choose a and b to minimize this so-called *mean square error*. Clearly, $E(Y - aX - b)^2$ is minimized, for any a,by choosing $b = E(Y - aX) = E(Y) - aE(X)$. With this choice of b, we find a such that

$$\begin{aligned} E(Y - aX - b)^2 =& E\{(Y - EY) - a(X - EX)\}^2 \\ =& \sigma_Y^2 - 2a\mu_{11} + a^2\sigma_X^2, \qquad \text{where } \sigma_X^2 = Var(X), \sigma_Y^2 = Var(Y), \mu_{11} = Cov(X,Y) \end{aligned}$$

is minimum. An easy computation shows that the minimum occurs if we choose $a = \frac{\mu_{11}}{\sigma_X^2}$ provided $\sigma_X^2 > 0$. Moreover, we can write

$$\begin{aligned}\min_{a,b} E(Y - aX - b)^2 =& \min_{a,b} \{\sigma_Y^2 - 2a\mu_{11} + a^2\sigma_X^2\} \\ =& \frac{\sigma_Y^2 - \mu_{11}^2}{\sigma_X^2} \\ =& \sigma_Y^2 \Big\{1 - [\frac{\mu_{11}}{(\sigma_X \sigma_Y)}]^2\Big\}.\end{aligned}$$

Let us denote

$$\rho = \frac{\mu_{11}}{\sigma_X \sigma_Y}.$$

Since $E(Y - aX - b)^2 \geq 0$,we must have $1 - \rho^2 \geq 0$, or $\rho^2 \leq 1$, or $|\rho| \leq 1$. In the case of equality, we have $|\rho| = 1$, or equivalently, $E(Y - aX - b)^2 = 0$ holds.This implies that $\mathbb{P}(Y = aX + b) = 1$. Here, for $a \neq 0$, we note that the signs of a and ρ are the same. Thus, if $\rho = 1$ then $\mathbb{P}(Y = aX + b)$ when $a > 0$, and if $\rho = -1$, then $a < 0$.

Alternate proof of Cauchy-Schwartz inequality: Random variables X and Y satisfy $\{E(XY)\} \leq E(X^2)E(Y^2)$ with equality holds iff $P(aX = bY) = 1$ for some a and b, at least one of which is non-zero.
Proof: We know from previous theorem that $|\rho(X, Y)| = |\frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}| \leq 1$, which implies

$$|\mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))]| \leq \sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]\mathbb{E}[(Y - \mathbb{E}[Y])^2]} \tag{3}$$

Let us assume two other random variables U and V with means $\mu_1,\ \mu_2$ and variances $\sigma_1^2,\ \sigma_2^2$, respectively. Now, we consider $X = \frac{U-\mu_1}{\sigma_1}$ and $Y = \frac{V-\mu_2}{\sigma_2}$. Therefore, $E[X] = 0$ and $E[Y] = 0$ and $Var(X) = E[X^2]$ and $Var(Y) = E[Y^2]$. As a result of this assumption, above inequality (3) becomes

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]} \tag{4}$$

Hence, we have the proof.

3.1 Examples of Discrete Random Variables

Bernoulli Random Variable: X is a Bernoulli random variable, if it takes the value 1(success) with probability p and 0(failure) with probability $q = 1 - p$. Then

$$f(1) = p,\ f(0) = 1 - p,$$

$$\mathbb{E}(X) = \sum_x x f(x) = 1 \cdot p + 0 \cdot q = p,$$

$$\mathbb{E}(X^2) = \sum_x x^2 f(x) = 1 \cdot p + 0 \cdot q = p,$$

$$Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = p^2 - p = pq.$$

Binomial Random Variable: The binomial random variable X counts the number of success in n Bernoulli trials. We write it as $X \sim Bin(n, p)$. Then the pmf is given by

$$f(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k}, \text{ where } k = 0, 1, 2 \ldots, n$$

and hence

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} \\
&= np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!((n-1)-(k-1))!} p^{k-1} (1-p)^{(n-1)-(k-1)} \\
&= np \sum_{k=0}^{n-1} \frac{n!}{k!(n-k)!} p^k (1-p)^{n-1-k} \\
&= np(p + 1 - p)^{n-1} \\
&= np, \\
\mathbb{E}(X^2) &= \sum_{k=0}^{n} k^2 \mathbb{P}(X = k) = \sum_{k=0}^{n} (k + (k-1)k) \mathbb{P}(X = x) \\
&= \sum_{k=0}^{n} k \mathbb{P}(X = k) + \sum_{k=0}^{n} k(k-1) \mathbb{P}(X = k) \\
&= np + \sum_{k=0}^{n} k(k-1) \binom{n}{k} p^k (1-p)^{n-k} \\
&= np + \sum_{k=2}^{n} k(k-1) \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\
&= np + n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!((n-2)-(k-2))!} p^{k-2} (1-p)^{(n-2)-(k-2)} \\
&= np + n(n-1)p^2 \sum_{k=0}^{n-2} \frac{n!}{k!(n-k)!} p^k (1-p)^{(n-2)-k} \\
&= np + n(n-1)p^2 (p + 1 - p)^{n-2} = np + n(n-1)p^2 \\
&= npq + n^2 p^2, \\
Var(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = npq.
\end{aligned}$$

Example 3.1. Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed Bernoulli random variable, then their sum $X = X_1 + X_2 + \ldots + X_n$ is identical to Binomial

random variable with parameter (n, p).

$$\mathbb{E}(X) = \mathbb{E}(X_1 + X_2 + \ldots + X_n) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \ldots + \mathbb{E}(X_n) = p + \ldots + p = np,$$
$$Var(X) = Var(X_1+X_2+\ldots+X_n) = Var(X_1)+Var(X_2)+\ldots+Var(X_n) = pq+\ldots+pq = npq.$$

Poisson Random Variable: The Poisson random variable X counts the number of success in infinite independent Bernoulli trials. Then for $\lambda > 0$, the pmf is given by

$$f(k) = \frac{\lambda^k}{k!}e^{-\lambda},\ k = 0, 1, 2, \ldots,$$

and hence

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=0}^{\infty} k\frac{e^{-\lambda}\lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda}\lambda^k}{(k-1)!} = \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda}e^{\lambda} = \lambda,\\
\mathbb{E}(X^2) &= \sum_{k=0}^{\infty} k^2\mathbb{P}(X = k) = \sum_{k=0}^{\infty}(k + (k-1)k)\mathbb{P}(X = k)\\
&= \sum_{k=0}^{\infty} k\mathbb{P}(X = k) + \sum_{k=0}^{\infty} k(k-1)\mathbb{P}(X = k) = \lambda + \sum_{k=2}^{\infty} k(k-1)\frac{e^{-\lambda}\lambda^k}{k!}\\
&= \lambda + \lambda^2 e^{-\lambda}\sum_{k=2}^{\infty}\frac{\lambda^{k-2}}{(k-2)!} = \lambda + \lambda^2 e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \lambda + \lambda^2 \cdot e^{-\lambda} \cdot e^{\lambda}\\
&= \lambda + \lambda^2,\\
Var(X) &= \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.
\end{aligned}$$

Poisson as an approximation to the binomial:
X is a binomial random variable with parameters (n, p), and let $\lambda = np$. Then

$$\begin{aligned}
f(k) = \mathbb{P}(X = k) &= \lim_{\substack{n\to\infty\\ np<\infty}} \binom{n}{k} p^k(1-p)^{n-k} = \lim_{\substack{np=\lambda\\ n\to\infty\\ p\to 0}} \binom{n}{k} p^k(1-p)^{n-k}\\
&= \frac{n(n-1)\ldots(n-k+1)}{n^k}\frac{\lambda^k}{k!}\frac{(1-\lambda/n)^n}{(1-\lambda/n)^k}.
\end{aligned}$$

Now, for large n and moderate λ, $(1-\lambda/n)^n \approx e^{-\lambda}$, $(1-\lambda/n)^k \approx 1$, $\frac{n(n-1)\ldots(n-k+1)}{n^k} \approx 1$.
Hence,

$$\mathbb{P}(X = k) \approx \frac{e^{-\lambda}}{k!}.$$

Geometric Random Variable: The Geometric random variable counts the number of independent Bernoulli trials required to get the first success. The pmf is given by

$$f(k) = \mathbb{P}(X = k) = (1-p)^{k-1}p,\ k = 1, 2, 3, \ldots$$

and

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=1}^{\infty} k(1-p)^{k-1}p = p\sum_{k=1}^{\infty} k(1-p)^{k-1} \\
&= p\left[\sum_{k=1}^{\infty}(1-p)^{k-1} + \sum_{k=2}^{\infty}(1-p)^{k-1} + \sum_{k=3}^{\infty}(1-p)^{k-1} + \dots\right] \\
&= p\left[\frac{1}{p} + \frac{(1-p)}{p} + \frac{(1-p)^2}{p} + \frac{(1-p)^3}{p} + \dots\right] \\
&= 1 + (1-p) + (1-p)^2 + (1-p)^3 + \dots = \frac{1}{p}, \\
\mathbb{E}(X(X-1)) &= \sum_{k=1}^{\infty} k(k-1)p(1-p)^{k-1} = p(1-p)\sum_{k=2}^{\infty} k(k-1)(1-p)^{k-2} \\
&= p(1-p)\sum_{k=2}^{\infty}\frac{d^2}{dp^2}(1-p)^k = p(1-p)\frac{d^2}{dp^2}\sum_{k=2}^{\infty}(1-p)^k \\
&= p(1-p)\frac{d^2}{dp^2}\left(\frac{(1-p)^2}{p}\right) = p(1-p)\frac{2}{p^3} = \frac{2(1-p)}{p^2}.
\end{aligned}$$

The interchange of summation and differentiation is justified by the fact that convergent power series converge uniformly on compact subsets of the set of points where they converge. And hence,

$$\begin{aligned}
Var(X) &= \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \mathbb{E}(X(X-1)) + \mathbb{E}(X) - \mathbb{E}(X)^2 \\
&= \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}.
\end{aligned}$$

Negative Binomial Random Variable: The Negative random variable X counts the number of Bernoulli trials k required to get r success. Thus, $X = X_1 + X_2 + \ldots + X_r$ where $X_i \overset{D}{\sim} Geom(p)$. Then the pmf is given by

$$f(k) = \mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \ k = r, r+1, \dots$$

and

$$\mathbb{E}(X) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \ldots + \mathbb{E}(X_r) = \frac{r}{p},$$

$$Var(X) = rVar(X_1) = \frac{r(1-p)}{p^2}.$$

Hyper Geometric Random Variable: The Hyper geometric random variable counts the number of success of drawing red balls from a bag of N balls containing K red balls, in n draws without replacement. Then the pmf will be given by

$$f(k) = \mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \ k = 0, 1, \dots, K,$$

and

$$\mathbb{E}[X] = \sum_{k=0}^{K} k \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}} = \sum_{k=1}^{K} \frac{k\binom{K}{k}\binom{N-K}{n-k}}{\frac{N}{n}\binom{N-1}{n-1}} = \sum_{k=1}^{K} \frac{K\binom{K-1}{k-1}\binom{N-K}{n-k}}{\frac{N}{n}\binom{N-1}{n-1}}$$
$$= \frac{nK}{N} \sum_{k=1}^{K} \frac{\binom{K-1}{k-1}\binom{N-K}{n-k}}{\binom{N-1}{n-1}} = \frac{nK}{N} \sum_{k=0}^{K-1} \frac{\binom{K-1}{k}\binom{(N-1)-(K-1)}{(n-1)-k}}{\binom{N-1}{n-1}}.$$

The term in summation above is the probability of getting k red balls when $n-1$ balls are drawn from a bag of $N-1$ balls containing $K-1$ red balls. Thus, the sum of these terms over $k = 0, 1, \ldots, K-1$ equals 1. Hence,

$$\mathbb{E}[X] = n\frac{K}{N}.$$

Also,

$$\mathbb{E}[X(X-1)] = \sum_{k=0}^{K} k(k-1) \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}} = \sum_{k=2}^{K} \frac{k\binom{K}{k}\binom{N-K}{n-k}}{\frac{N(N-1)}{n(n-1)}\binom{N-2}{n-2}}$$
$$= \sum_{k=2}^{K} \frac{K(K-1)\binom{K-2}{k-2}\binom{N-K}{n-k}}{\frac{N(N-2)}{n(n-2)}\binom{N-2}{n-2}} = \frac{n(n-1)K(K-1)}{N(N-1)} \sum_{k=2}^{K} \frac{\binom{K-2}{k-2}\binom{N-K}{n-k}}{\binom{N-2}{n-2}}$$
$$= \frac{n(n-1)K(K-1)}{N(N-1)} \sum_{k=0}^{K-2} \frac{\binom{K-2}{k}\binom{(N-2)-(K-2)}{(n-2)-k}}{\binom{N-2}{n-2}}.$$

The term in summation above is the probability of getting k red balls when $n-2$ balls are drawn from a bag of $N-2$ balls containing $K-2$ red balls. Thus, the sum of these terms over $k = 0, 1, \ldots, K-2$ equals 1. Hence,

$$\mathbb{E}[X(X-1)] = \frac{n(n-1)K(K-1)}{N(N-1)}.$$

Thus,

$$Var(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2$$
$$= \frac{n(n-1)K(K-1)}{N(N-1)} + \frac{nK}{N} - \left(\frac{nK}{N}\right)^2$$
$$= \frac{nK(N-n)(N-K)}{N^2(N-1)}.$$

3.2 Joint Distribution Function

The random variable X and Y on the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ are called **(jointly) discrete** if the random vector (X, Y) takes values in some countable subset of $\mathbb{R}^2$ only.

Definition 3.8 (Joint Distribution Function)**.** The **joint distribution function** $F : \mathbb{R}^2 \to [0, 1]$ of discrete random variables X and Y, is given by

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

Definition 3.9 (Joint Mass function)**.** The joint discrete random variables X, Y have **joint probability mass function** $f : \mathbb{R}^2 \to [0, 1]$ given by

$$f_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

This gives $F_{X,Y}(a, b) = \sum_{x=-\infty}^{a} \sum_{y=-\infty}^{b} f_{X,Y}(x, y)$.

Theorem 3.7. The discrete random variables X and Y are independent if and only if

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \ \forall \ x, y \in \mathbb{R}.$$

Proof: The proof is obvious by definition of independent random variables.

Definition 3.10 (Marginal Distribution Function)**.** Given a joint distribution function $F_{X,Y}$ of random variables X and Y, the **marginal distribution function** of X and Y are given by

$$F_X(x) = \mathbb{P}(X \leq x) = F_{X,Y}(x, \infty),$$
$$F_Y(y) = \mathbb{P}(Y \leq y) = F_{X,Y}(\infty, y).$$

Definition 3.11 (Marginal Mass Function)**.** Given a joint mass function $f_{X,Y}$ of random variables X and Y, the **marginal mass function** of X and Y are given by

$$f_X(x) = \mathbb{P}(X = x) = \sum_y f_{X,Y}(x, y),$$

$$f_Y(y) = \mathbb{P}(Y = y) = \sum_x f_{X,Y}(x, y).$$

This gives $F_X(a) = \sum_{x=-\infty}^{a} \sum_y f_{X,Y}(x, y)$, and $F_Y(b) = \sum_{y=-\infty}^{b} \sum_x f_{X,Y}(x, y)$.

Example 3.2. Let the joint probability mass function of random variable X and Y is

$$f(x, y) = \frac{\alpha^x \beta^y}{x! y!} e^{-\alpha-\beta} \text{ for } x, y = 0, 1, 2, \ldots$$

where $\alpha, \beta > 0$. The marginal mass function of X is

$$f_X(x) = \sum_y f(x, y) = \frac{\alpha^x}{x!} e^{-\alpha} \sum_{y=0}^{\infty} \frac{\beta^y}{y!} e^{-\beta} = \frac{\alpha^x}{x!} e^{-\alpha},$$

and

$$f_Y(y) = \sum_x f(x, y) = \frac{\beta^y}{y!} e^{-\beta} \sum_{y=0}^{\infty} \frac{\alpha^x}{x!} e^{-\alpha} = \frac{\beta^y}{y!} e^{-\beta}.$$

Thus, X has the Poisson distribution with parameter α and Y has the Poisson distribution with parameter β. Clearly $f(x, y) = f_X(x) f_Y(y)$, whence X and Y are independent.

Example 3.3. Let X and Y be two random variables that count the no. of heads and no. tails in N number of independent Bernoulli trials, respectively. Then $X \overset{D}{\sim} Bin(N, p)$ and $Y \overset{D}{\sim} Bin(N, 1-p)$.
Thus,

$$\mathbb{P}(X = x) = \binom{N}{x} p^x (1-p)^{N-x}$$

and

$$\mathbb{P}(Y = y) = \binom{N}{y} (1-p)^y p^{N-y}.$$

Also, $\mathbb{P}(X = x, Y = y) = 0$ if $y \neq N - x$; but $\mathbb{P}(X = x), \mathbb{P}(Y = y)$ are never zero. Thus, X and Y are dependent random variables.

3.3 Conditional Distribution Function

Definition 3.12 (Conditional Distribution Function)**.** The **conditional distribution function** of Y given $X = x$, written $F_{Y|X}(\cdot|x)$, is defined by

$$F_{Y|X}(y|x) = \mathbb{P}(Y \leq y | X = x) \ \forall \ x \in \mathbb{R}$$

such that $\mathbb{P}(X = x) > 0$.

Definition 3.13 (Conditional Probability Mass Function)**.** The **conditional probability mass function** of Y given $X = x$, written $f_{Y|X}(\cdot|x)$, is defined by

$$f_{Y|X}(y|x) = \mathbb{P}(Y = y | X = x)$$

for any $x \in \mathbb{R}$ such that $\mathbb{P}(X = x) > 0$.

The conditional probability mass function is also written as

$$f_{Y|X}(y|x) = \mathbb{P}(Y = y | X = x) = \frac{\mathbb{P}(Y = y, X = x)}{\mathbb{P}(X = x)} = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

Definition 3.14 (Conditional Expectation)**.** The **expectation of conditional distribution** of Y given $X = x$, written as $\mathbb{E}(Y|X = x)$, is defined as

$$\mathbb{E}(Y|X = x) = \sum_y y f_{Y|X}(y|x).$$

Let $\psi(x) = \mathbb{E}(Y|X = x)$, then $\psi(X)$ is called the conditional expectation of Y given X, written as $\mathbb{E}(Y|X)$.

Theorem 3.8. The conditional expectation $\psi(X) = \mathbb{E}(Y|X)$ satisfies

$$\mathbb{E}(\mathbb{E}(Y|X)) = \mathbb{E}(Y).$$

Proof:

$$\mathbb{E}[\mathbb{E}(Y|X)] = \mathbb{E}[\sum_y y f_{Y|X}(y|x)] = \sum_x \sum_y y f_{Y|X}(y|x) f_X(x) = \sum_y y \sum_x f_{Y|X}(y|x) f_X(x)$$
$$= \sum_y y \sum_x \frac{f_{X,Y}(x,y)}{f_X(x)} f_X(x) = \sum_y y \sum_x f_{X,Y}(x,y) = \sum_y y f_Y(y) = \mathbb{E}[Y].$$

Example 3.4. A hen lays N eggs, where N has the Poisson distribution with parameter λ. Each egg hateched with probability $p(= 1-q)$ independently of the other eggs. Let K be the number of chicks. Find $\mathbb{E}(K|N), K,$ and $N|K$.
$N \overset{D}{\sim} Poisson(\lambda)$ and $K|N \overset{D}{\sim} Binomial(N,p)$; so $\mathbb{P}(N=n) = \frac{e^{-\lambda}\lambda^n}{n!}$ and $\mathbb{P}(K=k|N=n) = \binom{n}{k} p^k (1-p)^{n-k}$.

$$\mathbb{E}(K|N) = Np$$

and

$$\mathbb{E}(K) = \mathbb{E}[\mathbb{E}[K|N]] = \mathbb{E}[Np] = \lambda p.$$

$$f_{N|K}(n|k) = \mathbb{P}(N=n|K=k) = \frac{\mathbb{P}(N=n, K=k)}{\mathbb{P}(K=k)} = \frac{\mathbb{P}(K=k|N=n)\mathbb{P}(N=n)}{\mathbb{P}(K=k)}$$
$$= \frac{\binom{n}{k} p^k (1-p)^{n-k} \frac{e^{-\lambda}\lambda^n}{n!}}{\sum_{n=k}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} \frac{e^{-\lambda}\lambda^n}{n!}} = \frac{\binom{n}{k} p^k (1-p)^{n-k} \frac{e^{-\lambda}\lambda^n}{n!}}{\frac{e^{-\lambda}p^k\lambda^k}{k!} \sum_{n=k}^{\infty} \frac{[(1-p)\lambda]^{n-k}}{(n-k)!}}$$
$$= \frac{\binom{n}{k} p^k (1-p)^{n-k} \frac{e^{-\lambda}\lambda^n}{n!}}{\frac{e^{-\lambda}p^k\lambda^k}{k!} e^{\lambda(1-p)}} = \frac{\binom{n}{k} p^k (1-p)^{n-k} \frac{e^{-\lambda}\lambda^n}{n!}}{e^{-\lambda p} \frac{(\lambda p)^k}{k!}} = \frac{e^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!},$$

where $n = k, k+1, \ldots$.

$$\mathbb{E}(N|K) = \sum_{n=k}^{\infty} \frac{n e^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!} = \sum_{n=k}^{\infty} \frac{(n-k+k) e^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!}$$
$$= \sum_{n=k+1}^{\infty} \frac{e^{-\lambda q}(\lambda q)^{n-k}}{(n-k-1)!} + k \sum_{n=k}^{\infty} \frac{e^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!}$$
$$= \lambda q \sum_{n=k+1}^{\infty} \frac{e^{-\lambda q}(\lambda q)^{n-k-1}}{(n-k-1)!} + k = k + \lambda q.$$

Thus, $\mathbb{E}(N|K) = k + \lambda q$.

Theorem 3.9. For random variables X and Y, and $i, j \in \mathbb{N}$,

$$\mathbb{E}[X^i Y^j] = \mathbb{E}_X[X^i \mathbb{E}_{Y|X}(Y^j|X)].$$

Proof: The statement can be proved as below,

$$\mathbb{E}[X^i Y^j] = \sum_{i,j} x^i y^j f_{X,Y}(i,j) = \sum_i x^i \Big[\sum_j y^j f_{Y|X}(j|X=i)\Big] f_X(i)$$
$$= \sum_i x^i \mathbb{E}_{Y|X}(Y^j|X) f_X(i) = \mathbb{E}_X[X^i \mathbb{E}_{Y|X}[Y^j|X]].$$

Theorem 3.10. Prove that, for random variables X and Y,

$$Var(Y) = E(Var(Y|X)) + Var(E(Y|X)).$$

Proof: We know that

$$\begin{aligned} Var(Y) &= E(Y^2) - [E(Y)]^2, \\ Var(Y|X) &= E(Y^2|X) - [E(Y|X)]^2. \end{aligned}$$

Then,

$$\begin{aligned} E[Var(Y|X)] &= E[E(Y^2|X)] - E[\{E(Y|X)\}^2] \\ &= E(Y^2) - E[\{E(Y|X)\}^2] \\ Var[E(Y|X)] &= E[\{E(Y|X)\}^2] - [E[E[(Y|X)]]^2 \\ &= E[\{E(Y|X)\}^2] - [E(Y)]^2. \end{aligned}$$

Adding the last two equations, we get

$$Var(Y) = E(Y^2) - [E(Y)]^2 = E(Var(Y|X)) + Var(E(Y|X)).$$

Theorem 3.11 (**Mutual Information**)**.** Let X and Y be discrete random variables with joint mass function f. Show that the **mutual information**

$$I = \mathbb{E}\left[log\left(\frac{f(X,Y)}{f_X(X)f_Y(Y)}\right)\right]$$

satisfies $I \geq 0$, with equality if and only if X and Y are independent.
Proof: We know that $log(y) \leq y - 1$ with equality if and only if $y = 1$. Thus, $-log(y) \geq 1 - y$ implies

$$\begin{aligned} I &= \mathbb{E}\left[log\left(\frac{f(X,Y)}{f_X(X)f_Y(Y)}\right)\right] = \mathbb{E}\left[-log\left(\frac{f_X(X)f_Y(Y)}{f(X,Y)}\right)\right] \\ &\geq \mathbb{E}\left[1 - \frac{f_X(X)f_Y(Y)}{f(X,Y)}\right] = 1 - \mathbb{E}\left[\frac{f_X(X)f_Y(Y)}{f(X,Y)}\right] \\ &= 1 - \sum_x \sum_y \left(\frac{f_X(x)f_Y(y)}{f(x,y)}\right) f(x,y) = 1 - \sum_x f(x) \sum_y f(y) \\ &= 1 - 1 \cdot 1 = 0. \end{aligned}$$

Hence $I \geq 0$, and equality holds iff $f(X,Y) = f_X(X)f_Y(Y)$ iff X and Y are independent.

Example 3.5. The lifetime of a machine (in days) is a random variable T with mass function f. Given that that the machine is working after t days, what is the mean subsequent lifetime of the machine when

(a) $f(x) = \frac{1}{N+1},\ x = 0, 1, 2, \ldots, N$

(b) $f(x) = \frac{1}{2^x},\ x = 1, 2, \ldots$

Example 3.6. Let $X_1 \overset{D}{\sim} Geometric(p),\ X_2 \overset{D}{\sim} Geometric(p)$ and X_1, X_2 are independent. Then

$$\begin{aligned} f_{X_1+X_2}(z) &= \sum_x f_{X_1}(x) f_{X_2}(z-x) = \sum_{x=1}^{z-1} p(1-p)^{x-1} p(1-p)^{z-x-1} \\ &= p^2 \sum_{x=1}^{z-1} (1-p)^{z-2} = \frac{p^2}{(1-p)^2}(1-p)^z \sum_{x=1}^{z-1} 1 \\ &= (z-1)p^2(1-p)^{z-2}. \end{aligned}$$

4 Continuous Random Variables

Definition 4.1 (Probability Density Function)**.** A random variable X is **continuous** if its distribution function $F(x) = \mathbb{P}(X \leq x)$ can be written as

$$F(x) = \int_{-\infty}^{x} f(u)du$$

for some integrable $f : \mathbb{R} \to [0, \infty)$. The function f is called the **(probability) density function** of the continuous random variable X.

Theorem 4.1. If a continuous random variable X has density function f then,

(a) $\int_{-\infty}^{\infty} f(x)dx = 1$,

(b) $\mathbb{P}(X = x) = 0 \ \forall \ x \in \mathbb{R}$,

(c) $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)dx$.

Proof: (a) $\int_{-\infty}^{\infty} f(x)dx = F(\infty) = \mathbb{P}(X \leq \infty) = 1$.
(b) By theorem 2.2,

$$\mathbb{P}(X = x) = F(x) - \lim_{y \uparrow x} F(y) = \int_{-\infty}^{x} f(u)du - \lim_{y \uparrow x} \int_{-\infty}^{y} f(u)du = \lim_{y \uparrow x} \int_{y}^{x} f(u)du = 0.$$

(c) By theorem 2.2, for $a < b$,

$$\begin{aligned}\mathbb{P}(a \leq X \leq b) &= \mathbb{P}(X = a) + \mathbb{P}(a < X \leq b) = 0 + F(b) - F(a) \\ &= \int_{-\infty}^{b} f(u)du - \int_{-\infty}^{a} f(u)du = \int_{a}^{b} f(u)du.\end{aligned}$$

Definition 4.2 (Independent Continuous Random Variable)**.** Two continuous random variables are called **independent** if $\{X \leq x\}$ and $\{Y \leq y\}$ are independent events for all $x, y \in \mathbb{R}$, that is,

$$\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x)\mathbb{P}(Y \leq y).$$

Definition 4.3 (Expectation)**.** The **expectation** of a continuous random variable X with pdf f is given by

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x)dx$$

whenever this integral exists.

Theorem 4.2. If f and g are density functions of random variables X and Y, show that for $0 \leq \alpha \leq 1$, $\alpha f + (1 - \alpha)g$ is a density function. And describe a random variable of which it is the density function.

Proof: Let $F_X(x) = \int_{-\infty}^{x} f(u)du$ and $G_Y(y) = \int_{-\infty}^{y} g(u)du$ be the distribution function of X and Y. Define a function $F : \mathbb{R} \to \mathbb{R}$ by

$$F(x) = \int_{-\infty}^{x} (\alpha f + (1-\alpha)g)(u)du = \alpha \int_{-\infty}^{x} f(u)du + (1-\alpha) \int_{-\infty}^{\infty} g(u)du = \alpha F_X(x) + (1-\alpha)G_Y(x).$$

Then by Example 2.2, F is a distribution function of some random variable, Z, with density function $\alpha f + (1-\alpha)g$. The random variable Z, takes value in X with probability α and value in Y with probability $(1-\alpha)$.

Example 4.1. Let X be a positive random variable with the density function f and distribution function F. Define the hazard function $H(x) = -log[1 - F(x)]$ and the hazard rate

$$r(x) = \lim_{h \downarrow 0} \frac{\mathbb{P}(X \leq x + h | X > x)}{h}, \ x \geq 0.$$

Prove that $r(x) = H'(x) = \frac{f(x)}{1-F(x)}$.

$$\begin{aligned} r(x) &= \lim_{h \downarrow 0} \frac{\mathbb{P}(X \leq x + h | X > x)}{h} = \lim_{h \downarrow 0} \frac{\mathbb{P}(X \leq x + h, X > x)}{h\mathbb{P}(X > x)} \\ &= \lim_{h \downarrow 0} \frac{\mathbb{P}(x < X \leq x + h)}{h(1 - F(x))} = \lim_{h \downarrow 0} \frac{f(x)h}{h(1 - F(x))} = \frac{f(x)}{1 - F(x)}. \end{aligned}$$

Definition 4.4 ($k^{th}-$ moment)**.** If k is a positive integer, the kth **moment** m_k of X is defined to be

$$m_k = \mathbb{E}(X^k) = \int_{-\infty}^{\infty} x^k f(x)dx.$$

The kth **central moment** σ_k is given by

$$\sigma_k = \int_{-\infty}^{\infty} (x - m_1)^k f(x)dx.$$

Theorem 4.3. If X has density function f with $f(x) = 0$ when $x < 0$, and distribution function F, then

$$\mathbb{E}(X) = \int_{0}^{\infty} [1 - F(x)]dx.$$

Proof:

$$\begin{aligned} \int_{0}^{\infty} [1 - F(x)]dx &= \int_{0}^{\infty} \mathbb{P}(X > x)dx = \int_{0}^{\infty} \int_{x}^{\infty} f(y)dydx \\ &= \int_{0}^{\infty} f(y) \int_{0}^{y} dxdy = \int_{0}^{\infty} yf(y)dy = \mathbb{E}(X). \end{aligned}$$

4.1 Examples of Continuous Random Variable

Uniform Distribution: The random variable X is said to have uniform distribution on $[a, b]$ if it has density function

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & otherwise. \end{cases}$$

$$\mathbb{E}(X) = \int_{-\infty}^{a} x \cdot 0 dx + \int_{a}^{b} x \frac{1}{b-a} dx + \int_{b}^{\infty} x \cdot 0 dx = \frac{x^2}{2(b-a)}\bigg|_a^b = \frac{a+b}{2},$$

$$\begin{aligned} Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 &= \int_{-\infty}^{\infty} x^2 f(x) dx - \left(\frac{a+b}{2}\right)^2 \\ &= \int_a^b x^2 \frac{1}{b-a} dx - \left(\frac{a+b}{2}\right)^2 = \frac{x^3}{3(b-a)}\bigg|_a^b - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{(b-a)^2}{12}. \end{aligned}$$

Exponential Distribution: The random variable X is said to have an exponential distribution with parameter $\lambda(> 0)$ if it has density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

The cdf of exponential r.v. is given by,

$$F(x) = 1 - e^{-\lambda x}, \ x \geq 0.$$

And

$$\mathbb{E}(X) = \int_0^{\infty} x \lambda e^{-\lambda x} = [-x e^{-\lambda x}]_0^{\infty} - \int_0^{\infty} (-e^{-\lambda x}) dx = 0 - \left[\frac{e^{-\lambda x}}{\lambda}\right]_0^{\infty} = \frac{1}{\lambda},$$

$$\mathbb{E}[X^2] = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = [-x^2 e^{-\lambda x}]_0^{\infty} - \left[2x \frac{e^{-\lambda x}}{\lambda}\right]_0^{\infty} + \left[-2 \frac{e^{-\lambda x}}{\lambda^2}\right]_0^{\infty} = \frac{2}{\lambda^2},$$

$$Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}.$$

Normal Distribution: The random variable X is said to have normal distribution with parameters μ and σ^2, if it has density function

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp - \frac{(x-\mu)^2}{2\sigma^2}, \ \infty < x < \infty.$$

It is denoted by $N(\mu, \sigma^2)$. The expectation of the normal random variable is given by

$$
\begin{aligned}
\mathbb{E}[X] &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (\sqrt{2}\sigma t + \mu) e^{-t^2} \sqrt{2}\sigma dt \qquad \left[\text{Put } \frac{x-\mu}{\sqrt{2}\sigma} = t\right] \\
&= \frac{\sqrt{2}\sigma}{\sqrt{\pi}} \int_{-\infty}^{\infty} t e^{-t^2} dt + \frac{\mu}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt \\
&= 0 + \frac{\mu}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt \\
&= \frac{2\mu}{\sqrt{\pi}} \int_{0}^{\infty} e^{-t^2} dt \\
&= \frac{2\mu}{\sqrt{\pi}} \int_{0}^{\infty} \frac{e^{-v}}{2\sqrt{v}} dv \qquad [\text{Put } t^2 = v] \\
&= \frac{\mu}{\sqrt{\pi}} \int_{0}^{\infty} v^{\frac{1}{2}-1} e^{-v} dv \\
&= \frac{\mu}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) \\
&= \frac{\mu}{\sqrt{\pi}} \cdot \sqrt{\pi} \\
&= \mu,
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}(X^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x^2 e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (\sqrt{2}\sigma t + \mu)^2 e^{-t^2} \sqrt{2}\sigma dt \qquad [\text{Put } \frac{x-\mu}{\sqrt{2}\sigma} = t] \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2} dt + \frac{\mu^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt + \frac{2\sqrt{2}\sigma\mu}{\sqrt{\pi}} \int_{-\infty}^{\infty} t e^{-t^2} dt \\
&= \frac{4\sigma^2}{\sqrt{\pi}} \int_{0}^{\infty} t^2 e^{-t^2} dt + \frac{2\mu^2}{\sqrt{\pi}} \int_{0}^{\infty} e^{-t^2} dt + 0 \\
&= \frac{4\sigma^2}{\sqrt{\pi}} \int_{0}^{\infty} \frac{u^{\frac{3}{2}-1}}{2} e^{-u} du + \frac{2\mu^2}{\sqrt{\pi}} \int_{0}^{\infty} \frac{u^{\frac{1}{2}-1}}{2} e^{-u} du \qquad [\text{Put } t^2 = u] \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) + \frac{\mu^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2} + 1\right) + \frac{\mu^2}{\sqrt{\pi}} \cdot \sqrt{\pi} \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) + \mu^2 \qquad [\because \Gamma(z+1) = z\Gamma(z)] \\
&= \frac{\sigma^2}{\sqrt{\pi}} \cdot \sqrt{\pi} + \mu^2 \\
&= \sigma^2 + \mu^2.
\end{aligned}
$$

The variance of normal random variable is given by

$$Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \sigma^2 + \mu^2 - \mu^2 = \sigma^2.$$

If $\mu = 0, \sigma^2 = 1$, then this distribution is called as **standard normal distribution**. And it is denoted by $N(0,1)$.

Example 4.2. If $X \overset{D}{\sim} N(\mu, \sigma^2)$, then $Y = \frac{X-\mu}{\sigma}$ follows standard normal distribution, i.e., $Y \overset{D}{\sim} N(0,1)$.
The cdf of Y will be given by

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) = \mathbb{P}(\frac{X-\mu}{\sigma} \leq y) \\ &= \mathbb{P}(X \leq \mu + \sigma y) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\mu+\sigma y} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \\ &= \frac{1}{2\pi\sigma^2} \int_{-\infty}^{y} e^{-\frac{t^2}{2}} \sigma dt \qquad\qquad [\text{Put } \frac{x-\mu}{\sigma} = t] \end{aligned}$$

The density function of Y can be obtained from cdf as follows:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \qquad -\infty < y < \infty.$$

Hence, $Y \overset{D}{\sim} N(0,1)$.

Gamma Distribution: The random variable X is said to have gamma distribution with parameters $t, \lambda > 0$, if it has density

$$f(x) = \frac{1}{\Gamma(t)} \lambda^t x^{t-1} e^{-\lambda x}, x \geq 0.$$

Here, $\Gamma(t)$ is the gamma function

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx.$$

If $t = 1$, $\Gamma(\lambda, t) \overset{D}{\sim} exp(\lambda)$.
If $\lambda = 1/2$ and $t = d/2$, then $\Gamma(\lambda, t) \overset{D}{\sim} \chi^2(d)$.
If $X \overset{D}{\sim} N(0,1)$, then $X^2 \overset{D}{\sim} \chi^2(1)$.
If $X, Y \overset{D}{\sim} N(0,1)$, then $X^2 + Y^2 \overset{D}{\sim} \chi^2(2)$.
If $X_1, X_2, \ldots, X_n \overset{D}{\sim} N(0,1)$, then $\sum_{i=1}^n X_i^2 \overset{D}{\sim} \chi^2(n)$.
The expectation and variance of gamma distribution is given by

$$\begin{aligned} \mathbb{E}[X] &= \int_0^\infty \frac{x}{\Gamma(t)} \lambda^t x^{t-1} e^{-\lambda x} dx = \frac{\lambda^t}{\Gamma(t)} \int_0^\infty \left(\frac{y}{\lambda}\right)^t e^{-y} \frac{dy}{\lambda} \qquad \left[\text{Put } x = \frac{y}{\lambda}\right] \\ &= \frac{1}{\lambda\Gamma(t)} \int_0^\infty y^{(t+1)-1} e^{-y} dy = \frac{\Gamma(t+1)}{\lambda\Gamma(t)} = \frac{t\Gamma(t)}{\lambda\Gamma(t)} \\ &= \frac{t}{\lambda}, \end{aligned}$$

$$
\begin{aligned}
\mathbb{E}[X^2] &= \int_0^\infty \frac{x^2}{\Gamma(t)}\lambda^t x^{t-1}e^{-\lambda x}dx = \frac{\lambda^t}{\Gamma(t)}\int_0^\infty \left(\frac{y}{\lambda}\right)^{t+1} e^{-y}\frac{dy}{\lambda} \qquad \left[\text{Put } x = \frac{y}{\lambda}\right] \\
&= \frac{1}{\lambda^2\Gamma(t)}\int_0^\infty y^{(t+2)-1}e^{-y}dy = \frac{\Gamma(t+2)}{\lambda^2\Gamma(t)} = \frac{t(t+1)\Gamma(t)}{\lambda^2\Gamma(t)} \\
&= \frac{t(t+1)}{\lambda^2},
\end{aligned}
$$

$$
Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{t(t+1)}{\lambda^2} - \left(\frac{t}{\lambda}\right)^2 = \frac{t}{\lambda^2}.
$$

Beta Distribution: The random variable X is said to have a Beta distribution with parameters $a, b > 0$, if it has density function

$$
f(x) = \frac{x^{a-1}(1-x)^{b-1}}{\int_0^1 x^{a-1}(1-x)^{b-1}dx}, \ 0 \le x \le 1.
$$

We denote this distribution by $\beta(a, b)$. The beta function

$$
B(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1}dx
$$

is related to Gamma function as $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

$$
\begin{aligned}
\mathbb{E}[X] &= \int_0^1 xf(x)dx = \int_0^1 x\frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}dx = \int_0^1 \frac{x^{(a+1)-1}(1-x)^{b-1}}{B(a,b)}dx \\
&= \frac{B(a+1,b)}{B(a,b)} = \frac{\Gamma(a+1)\Gamma(b)\Gamma(a+b)}{\Gamma(a+b+1)\Gamma(a)\Gamma(b)} = \frac{a\Gamma(a)\Gamma(b)\Gamma(a+b)}{(a+b)\Gamma(a+b)\Gamma(a)\Gamma(b)} \\
&= \frac{a}{a+b}, \\
\mathbb{E}[X^2] &= \int_0^1 x^2 f(x)dx = \int_0^1 x^2\frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}dx = \int_0^1 \frac{x^{(a+2)-1}(1-x)^{b-1}}{B(a,b)}dx \\
&= \frac{B(a+2,b)}{B(a,b)} = \frac{\Gamma(a+2)\Gamma(b)\Gamma(a+b)}{\Gamma(a+b+2)\Gamma(a)\Gamma(b)} = \frac{a(a+1)\Gamma(a)\Gamma(b)\Gamma(a+b)}{(a+b)(a+b+1)\Gamma(a+b)\Gamma(a)\Gamma(b)} \\
&= \frac{a(a+1)}{(a+b)(a+b+1)},
\end{aligned}
$$

$$
Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \left(\frac{a(a+1)}{(a+b)(a+b+1)}\right) - \left(\frac{a}{a+b}\right)^2 = \frac{ab}{(a+b)^2(a+b+1)}.
$$

If $a = b = 1$ then X is uniform on $[0, 1]$.

Cauchy Distribution: The random variable X has Cauchy distribution with parameter μ, θ if it has density function

$$
f(x) = \frac{\mu}{\pi}\frac{1}{(\mu^2 + (x-\theta)^2)}, \ -\infty < x < \infty, \mu > 0.
$$

We write $X \overset{D}{\sim} \mathscr{C}(\mu, \theta)$. Substituting $y = \frac{x-\theta}{\mu}$ we get

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{\infty} \frac{\mu}{\pi} \frac{1}{(\mu^2 + (x-\theta)^2)} dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1}{1+y^2} dy$$
$$= \frac{2}{\pi} \int_{0}^{\infty} \frac{1}{1+y^2} dy = \frac{2}{\pi} [tan^{-1}(y)]_0^{\infty} = 1.$$

The distribution function of $\mathscr{C}(\mu, \theta)$ is given by

$$F(x) = \int_{-\infty}^{x} f(u)du = \int_{-\infty}^{x} \frac{\mu}{\pi} \frac{1}{(\mu^2 + (u-\theta)^2)} du = \frac{1}{\pi} \int_{-\infty}^{\frac{x-\theta}{\mu}} \frac{1}{1+y^2} dy$$
$$= \frac{1}{\pi} [tan^{-1}(y)]_{-\infty}^{\frac{x-\theta}{\mu}} = \frac{1}{2} + \frac{1}{\pi} tan^{-1}\left(\frac{x-\theta}{\mu}\right).$$

The moment of order < 1 exist, but moments of order ≥ 1 does not exist for the random variable X. This can be seen as follows;

$$\mathbb{E}[|X|^{\alpha}] = \frac{\mu}{\pi} \int_{-\infty}^{\infty} x^{\alpha} \frac{1}{\mu^2 + (x-\theta)^2} dx = \frac{2}{\pi} \int_{0}^{\infty} \frac{x^{\alpha}}{1+x^2} dx$$
$$= \frac{1}{\pi} \int_{0}^{1} z^{\frac{(1-\alpha)}{2} - 1} (1-z)^{\frac{(1+\alpha)}{2} - 1} dz \qquad \left[\text{Put } z = \frac{1}{1+x^2}\right]$$

which converges for $\alpha < 1$ and diverges for $\alpha \geq 1$. An alternate proof of this property of moment is given in section 6.

Weibull Distribution: The random variable X is said to have Weibull distribution with parameters $\alpha, \beta > 0$, if it has the distribution function

$$F(x) = 1 - e^{-\alpha x^{\beta}}, \ x \geq 0.$$

By differentiation, we get the density function as

$$f(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}}, x \geq 0.$$

$$\mathbb{E}[X^k] = \int_{0}^{\infty} x^k \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}} dx = \int_{0}^{\infty} \frac{u^{(k/\beta+1)-1}}{\alpha^{k/\beta}} e^{-u} du$$
$$= \frac{\Gamma\left(\frac{k}{\beta} + 1\right)}{\alpha^{k/\beta}} = \frac{k\Gamma(k/\beta)}{\beta\alpha^{k/\beta}}. \quad [Put\ \alpha x^{\beta} = u].$$

Hence,

$$\mathbb{E}[X] = \frac{\Gamma(1/\beta)}{\beta\alpha^{1/\beta}},$$
$$Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2\Gamma(2/\beta)}{\beta\alpha^{2/\beta}} - \left(\frac{\Gamma(1/\beta)}{\beta\alpha^{1/\beta}}\right)^2$$
$$= \frac{\Gamma(\frac{2}{\beta} + 1) - \Gamma(1/\beta)^2}{\beta\alpha^{2/\beta}}.$$

If $\beta = 1$, it gives $exp(\alpha)$.

Exercise: Find the hazard rate for the following cases:

(a) $X \overset{D}{\sim} Weibull(\alpha, \beta)$,

(b) $X \overset{D}{\sim} exp(\lambda)$,

(c) X has density function $\alpha f + (1-\alpha)g$ where $0 < \alpha < 1$, and f and g are the densities of exponential variables with respective parameters λ, μ. What happens to this hazard rate $r(x)$ with limiting case $x \to \infty$?

Example 4.3. Let $X \overset{D}{\sim} N(0,1)$ and $Y = e^X$. Find the density function of Y. Name the random variable Y.

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(e^X \leq y) = \mathbb{P}(X \leq ln(y)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{ln(y)} e^{-\frac{x^2}{2}} dx.$$

Differentiate both sides w.r.t y,

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(ln(y))^2}{2}} \frac{1}{y} = \frac{1}{y\sqrt{2\pi}} e^{-\frac{(ln(y))^2}{2}}, \quad y > 0.$$

4.2 Joint Distribution Function

Definition 4.5 (Joint Distribution Function)**.** The **joint distribution function** of X and Y is the function $F : \mathbb{R}^2 \to [0,1]$ given by

$$F(x,y) = \mathbb{P}(X \leq x, Y \leq y).$$

Definition 4.6 (Joint Density Function)**.** The random variables X and Y are **jointly continuous** with **joint probability density function** $f : \mathbb{R}^2 \to [0, \infty)$ if

$$F(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u,v) du dv.$$

That is, if F is sufficiently differentiable, then usually

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y) = \frac{\partial^2}{\partial y \partial x} F(x,y).$$

Definition 4.7 (Marginal Distribution Function)**.** The **marginal distribution function** of X and Y are

$$F_X(x) = \mathbb{P}(X \leq x) = F(x, \infty),$$
$$F_Y(y) = \mathbb{P}(Y \leq y) = F(\infty, y).$$

Using above equations, we get

$$F_X(x) = \int_{u=-\infty}^{x} \int_{v=-\infty}^{\infty} f(u,v) dv du, \quad F_Y(y) = \int_{u=-\infty}^{\infty} \int_{v-\infty}^{y} f(u,v) dv du.$$

Definition 4.8 (Marginal Density Function)**.** The **marginal density function** of X and Y are given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy,$$
$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx.$$

If $g : \mathbb{R}^2 \to \mathbb{R}$ is a sufficiently nice function, then

$$\mathbb{E}[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)f(x,y)dxdy.$$

Theorem 4.4. The random variables X and Y are **independent** if and only if

$$F_{X,Y}(x,y) = F_X(x)F_Y(y),$$

which, for continuous random variable, is equivalent to

$$f_{X,Y}(x,y) = f_X(x)f_Y(y).$$

whenever F is differentiable at (x,y).
Proof: Suppose that X and Y are independent random variables, then

$$\begin{aligned} f_{X,Y}(x,y) &= \frac{\partial^2}{\partial x \partial y}F_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y}\mathbb{P}(X \le x, Y \le y) = \frac{\partial^2}{\partial x \partial y}\mathbb{P}(X \le x)\mathbb{P}(Y \le y) \\ &= \frac{\partial^2}{\partial x \partial y}F_X(x)F_Y(y) = f_X(x)f_Y(y). \end{aligned}$$

Conversely, if $f_{X,Y}(x,y) = f_X(x)f_Y(y)$, then for two Borel sets A and B

$$\begin{aligned} \mathbb{P}(A \in X, B \in Y) &= \int_A\int_B f_{X,Y}(x,y)dydx = \int_A\int_B f_X(x)(x)f_Y(y)dydx \\ &= \int_A f_X(x)dx \int_B f_Y(y)dy = \mathbb{P}(A \in X)\mathbb{P}(B \in Y). \end{aligned}$$

Bivariate Normal Distribution: A pair of random variables X and Y is said to have a bivariate normal distribution if their joint density function is defined as $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}}exp\left(-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)\right)$$

where ρ is a constant satisfying $-1 < \rho < 1$.

$$\begin{aligned} f(x,y) &= \frac{1}{2\pi\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}(x^2-2\rho xy+y^2)} \\ &= \frac{1}{2\pi\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}(x^2+\rho^2y^2-2\rho xy+y^2-\rho^2y^2)} \\ &= \frac{1}{2\pi\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}(x-\rho y)^2}e^{-\frac{y^2}{2}}. \end{aligned}$$

Thus, the marginal distribution of Y will be given by

$$f_Y(y) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x-\rho y)^2} e^{-\frac{y^2}{2}} dx = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}.$$

Similarly, $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.That is, $X, Y \overset{D}{\sim} N(0,1)$.
Hence, $\mathbb{E}[X] = 0 = \mathbb{E}[Y]$ and $Var(X) = 1 = Var(Y)$.

$$\begin{aligned}
\mathbb{E}[XY] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} x e^{-\frac{x^2}{2}} dx \int_{-\infty}^{\infty} y \frac{1}{\sqrt{2\pi(1-\rho^2)}} e^{-\frac{(y-\rho x)^2}{2(1-\rho^2)}} dy \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-\frac{x^2}{2}} \rho x dx \\
&= \rho \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx \\
&= \rho \cdot Var(X) \\
&= \rho.
\end{aligned}$$

Hence,

$$Cov(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \rho - 0 = \rho.$$

The pair of random variables $X \overset{D}{\sim} N(\mu_1, \sigma_1^2)$ and $Y \overset{D}{\sim} N(\mu_2, \sigma_2^2)$ with correlation ρ is said to have bivariate normal distribution if their joint density function is given by

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2}Q(x,y)},$$

where

$$Q(x,y) = \frac{1}{1-\rho^2}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right].$$

4.3 Conditional Distribution Function

The conditional distribution of random variable Y given $X = x$ will be

$$F(Y \le y | x \le X \le x + dx) = \frac{\mathbb{P}(Y \le y, x \le X \le x + dx)}{\mathbb{P}(x \le X \le x + dx)} = \frac{\int_{-\infty}^{y} f(x,v) dv dx}{f_X(x) dx} = \frac{\int_{-\infty}^{y} f(x,v) dv}{f_X(x)}$$

As $dx \downarrow 0$, the LHS of above equation approaches to our intuitive notion of the probability that $Y \le y$ given that $X = x$.

Definition 4.9 (Conditional Distribution Function)**.** The **conditional distribution function** of Y given $X = x$ is the function $F_{Y|X}(\cdot|x)$ given by

$$F_{Y|X}(y|x) = \int_{-\infty}^{y} \frac{f(x,v)}{f_X(x)} dv$$

for any x such that $f_X(x) > 0$.

Definition 4.10 (Conditional Density Function)**.** The **conditional density function** of $F_{Y|X}$, written $f_{Y|X}$, is given by

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

for any x such that $f_X(x) > 0$.

Definition 4.11 (Conditional Expectation)**.** The **conditional expectation** of Y given $X = x$ is defined as

$$\mathbb{E}[Y|X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy.$$

Theorem 4.5. The conditional expectation of Y given $X = x$ satisifies:

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y].$$

Proof: We can use the Fubini's theorem to get,

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[Y|X]] &= \mathbb{E}\left[\int_{y=-\infty}^{\infty} y f_{Y|X}(y|x) dy\right] = \int_{x=-\infty}^{\infty}\int_{y=-\infty}^{\infty} y f_{Y|X}(y|x) dy f_X(x) dx \\
&= \int_{y=-\infty}^{\infty} y \int_{x=-\infty}^{\infty} \frac{f_{X,Y}(x,y)}{f_X(x)} f_X(x) dx dy = \int_{y=-\infty}^{\infty} y \int_{x=-\infty}^{\infty} f_{X,Y}(x,y) dx dy \\
&= \int_{y=-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}[Y].
\end{aligned}$$

Example 4.4. For random variables X and Y having standard bivariate normal distribution, the joint density function is

$$f(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{(y-\rho x)^2}{2(1-\rho^2)}} e^{-\frac{x^2}{2}}$$

and the marginal distribution function is

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.$$

Thus, the conditional distribution of Y given X will be

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{1}{\sqrt{2\pi(1-\rho^2)}} e^{-\frac{(y-\rho x)^2}{2(1-\rho^2)}}.$$

That is, $Y|X \overset{D}{\sim} N(\rho x, 1-\rho^2)$. Hence, $E[Y|X] = \rho x$ and

$$E[XY] = E_X[X E_{Y|X}[Y]] = E_X[X\rho X] = \rho E_X[X^2] = \rho.$$

Example 4.5. Let X and Y have joint probability density function

(a) $f_{X,Y}(x,y) = \lambda^2 e^{-\lambda y};\ 0 \le x \le y < \infty,$

Find $E[Y|X]$, $Var(Y|X)$ and $Corr(X,Y)$ or $\rho(X,Y)$.
(a) We can write

$$f_X(x) = \int_x^\infty \lambda^2 e^{-\lambda y} dy = \lambda e^{-\lambda x}, \text{ where } x \ge 0,$$
$$f_Y(y) = \int_0^y \lambda^2 e^{-\lambda y} dx = \lambda^2 y e^{-\lambda y}, \text{ where } y \ge 0,$$
$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\lambda^2 e^{-\lambda y}}{\lambda e^{-\lambda x}} = \lambda e^{-\lambda(y-x)}, \text{ where } \infty > y \ge x.$$

Then,

$$E(Y|X) = \lambda \int_x^\infty y e^{-\lambda(y-x)} dy = x + \frac{1}{\lambda},$$
$$E[E(Y|X)] = E(X) + \frac{1}{\lambda} = \frac{2}{\lambda}, \text{ Since, } E[X] = \frac{1}{\lambda},\ E[X^2] = \frac{2}{\lambda^2},\ VarX = \frac{1}{\lambda^2}$$
$$E(Y) = \lambda^2 \int_0^\infty y^2 e^{-\lambda y} dy = \frac{2}{\lambda}.$$
$$E(Y^2|X) = \lambda \int_x^\infty y^2 e^{-\lambda(y-x)} dy = x^2 + \frac{2x}{\lambda} + \frac{2}{\lambda^2}$$
$$Var(Y|X) = E(Y^2|X) - [E(Y|X)]^2 = x^2 + \frac{2x}{\lambda} + \frac{2}{\lambda^2} - (x + \frac{1}{\lambda})^2 = \frac{1}{\lambda^2}.$$
$$E(Y^2) = E_X[E_{Y|X}[Y^2]] = E_X[X^2 + \frac{2X}{\lambda} + \frac{2}{\lambda^2}] = \frac{2}{\lambda^2} + \frac{2}{\lambda}E[X] + \frac{2}{\lambda^2} = \frac{6}{\lambda^2}$$
$$Var(Y) = E(Y^2) - [E[Y]]^2 = \frac{6}{\lambda^2} - \frac{4}{\lambda^2} = \frac{2}{\lambda^2}$$
$$E[XY] = E_X[XE_{Y|X}[Y]] = E_X[X(X + \frac{1}{\lambda})] = E[X^2 + X\frac{1}{\lambda}] = E[X^2] + \frac{1}{\lambda}E[X]$$

Thus, $E[XY] = \frac{2}{\lambda^2} + \frac{1}{\lambda^2} = \frac{3}{\lambda^2}$, $\rho(X,Y) = \frac{E[XY] - E[X]E[Y]}{\sqrt{Var(X)Var(Y)}} = \frac{\frac{3}{\lambda^2} - \frac{1}{\lambda}\frac{2}{\lambda}}{\sqrt{\frac{1}{\lambda^2}\frac{2}{\lambda^2}}} = \frac{1}{\sqrt{2}}.$

Example 4.6. Let the joint density function of two random variables X and Y be given by

$$f(x,y) = \frac{n!}{x!y!(n-x-y)!} p^x q^y (1-p-q)^{n-x-y},$$

where $p + q < 1, x, y = 0, 1, \ldots, n, x + y \le n$, then $f_X(x) = \frac{n!}{x!(n-x!)} p^x (1-p)^{n-x}$ and

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{\frac{n!}{x!y!(n-x-y)!} p^x q^y (1-p-q)^{n-x-y}}{\frac{n!}{x!(n-x!)} p^x (1-p)^{n-x}}$$
$$= \frac{(n-x)!}{x!(n-x)!} \frac{q^y}{(1-p)^y} \left(\frac{1-p-q}{1-p}\right)^{n-x-y}.$$

Thus, $Y|X \overset{D}{\sim} Bin(n-x, \frac{q}{1-p})$. This implies, $\mathbb{E}[Y|X] = \frac{(n-x)q}{1-p}$ and $Var(Y|X) = \frac{(n-x)q(1-p-q)}{(1-p)^2}$.
Also, $\mathbb{E}[\mathbb{E}[Y|X]] = \frac{(n-np)q}{1-p} = nq = \mathbb{E}[Y]$ and $Y \overset{D}{\sim} Binomial(n, q)$.

5 Memoryless Property

Definition 5.1 (Memoryless Property)**.** A random variable X is said to have a memoryless property if for any two times t, s, it satisfies

$$\mathbb{P}(X > t + s | X > s) = \mathbb{P}(X > t).$$

Theorem 5.1. A discrete random variable satisfies the memoryless property if and only if it is a Geometric random variable.
Proof: If $X \sim Geom(p)$ is a geometric random variable, then $\mathbb{P}(X = n) = p(1-p)^{n-1}$. Thus, for time n, k,

$$\begin{aligned}
\mathbb{P}(X > n + k | X > k) &= \frac{\mathbb{P}(X > n + k, X > k)}{\mathbb{P}(X > k)} = \frac{\mathbb{P}(X > n + k)}{\mathbb{P}(X > k)} \\
&= \frac{\sum_{i=n+k+1}^{\infty} p(1-p)^{i-1}}{\sum_{i=k+1}^{\infty} p(1-p)^{i-1}} = \frac{(1-p)^{n+k}}{(1-p)^k} \\
&= (1-p)^n = \mathbb{P}(X > n).
\end{aligned}$$

Thus, a geometric random variable satisfies the memoryless property.
Conversely, suppose a random variable X satifies the memoryless property. Then for time n, k, $\mathbb{P}(X > n + k | X > k) = \mathbb{P}(X > n)$. Putting $k = 1$ gives

$$\begin{aligned}
\mathbb{P}(X > n + 1) &= \mathbb{P}(X > n + 1, X > 1) = \mathbb{P}(X > n + 1 | X > 1)\mathbb{P}(X > 1) \\
&= \mathbb{P}(X > n)\mathbb{P}(X > 1) = \mathbb{P}(X > n - 1)\mathbb{P}(X > 1)^2 \\
&= \mathbb{P}(X > n - 2)\mathbb{P}(X > 1)^3 = \ldots = \mathbb{P}(X > 1)\mathbb{P}(X > 1)^n \\
&= \mathbb{P}(X > 1)^{n+1}.
\end{aligned}$$

Let $0 \leq p \leq 1$ defined as $p = 1 - \mathbb{P}(X > 1)$, then

$$\begin{aligned}
\mathbb{P}(X = n) &= \mathbb{P}(X > n - 1) - \mathbb{P}(X > n) = \mathbb{P}(X > 1)^{n-1} - \mathbb{P}(X > 1)^n \\
&= (1 - \mathbb{P}(X > 1))\mathbb{P}(X > 1)^{n-1} = p(1-p)^{n-1}.
\end{aligned}$$

Thus, X is a geometric random variable with parameter $p = 1 - \mathbb{P}(X > 1)$.

Theorem 5.2. A continuous random variable satisfies the memoryless property if and only if it is an Exponential random variable.
Proof: If $X \sim exp(\lambda)$ is an exponential random variable, then its density function is given by $f_X(x) = \lambda e^{-\lambda x}$ if $x > 0$ and 0 *otherwise*. The cumulative distribution function of X will be given by

$$F_X(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}.$$

Thus, for time t, s,

$$\begin{aligned}
\mathbb{P}(X > t + s | X > t) &= \frac{\mathbb{P}(X > t + s, X > t)}{\mathbb{P}(X > t)} = \frac{\mathbb{P}(X > t + s)}{\mathbb{P}(X > t)} \\
&= \frac{1 - F_X(t + s)}{1 - F_X(t)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} = e^{-\lambda t} = \mathbb{P}(X > t).
\end{aligned}$$

Thus, an exponential random variable satisfies the memoryless property.
Conversely, suppose a random variable X satifies the memoryless property. Then for time t, s, $\mathbb{P}(X > t + s | X > t) = \mathbb{P}(X > t)$. Let F be the cdf of X, and let $G(x) = \mathbb{P}(X > x) = 1 - F(x)$. The memoryless property says that $G(t + s) = G(s)G(t)$. Thus, for any $k \in \mathbb{N}$,

$$G(kt) = G((k-1)t + t) = G((k-1)t)G(t) = \ldots = G(t)^k,$$

$$G\left(\frac{t}{k}\right) = \left(G\left(\frac{t}{k}\right) \cdot G\left(\frac{t}{k}\right) \cdot \overset{k\ times}{\cdots} \cdot G\left(\frac{t}{k}\right)\right)^{\frac{1}{k}} = G\left(\frac{kt}{k}\right)^{\frac{1}{k}} = G(t)^{\frac{1}{k}}.$$

Thus, for any rational number $m/n \in \mathbb{Q}$,

$$G\Big(\frac{m}{n}t\Big) = G\Big(\frac{mt}{n}\Big) = G(mt)^{\frac{1}{n}} = G(t)^{\frac{m}{n}}.$$

Since every real number is the limit of some sequence of rational numbers, for any real $x > 0$, $G(xt) = G(t)^x$. Putting $t = 1$, we get $G(x) = G(1)^x = e^{xln(G(1))}$ to be an exponential function. Here $G(1) > 0$, because $0 \leq G(x) = \mathbb{P}(X \geq x) \leq 1$ and $G(x_0) > 0$ for some $x_0 \in \mathbb{R}$ implies $G(1) = G(x_0/x_0) = G(x_0)^{1/x_0}$. Thus, taking $ln(G(1)) = -\lambda$ for some $\lambda > 0$ implies $G(x) = e^{-\lambda x}$. And hence, X is an exponential random variable with parameter $\lambda = -ln(\mathbb{P}(X > 1))$.

6 Functions of Random Variables

Theorem 6.1. Let X be a random variable defined on $(\Omega, \mathscr{F}, \mathbb{P})$. Also, let g be a measurable function on $\mathbb{R}$. Then $g(X)$ is also a random variable.
Proof: For $y \in \mathbb{R}$, we have

$$\{g(X) \leq y\} = \{X \in g^{-1}(-\infty, y]\},$$

and since g is a measurable function, $g^{-1}(-\infty, y)$ is a measurable set. It follows that $\{g(X) \leq y\} \in \mathscr{F}$ and the proof is complete.

Theorem 6.2. Given a random variable X with a known distribution function, the distribution of the random variable $Y = g(X)$, where g is a measurable function is determined.
Proof: Indeed, for all $y \in \mathbb{R}$,

$$\mathbb{P}(\{Y \leq y\}) = \mathbb{P}(\{X \in g^{-1}(-\infty, y]\}).$$

In what follows, we will always assume that the functions under consideration are measurable.

Example 6.1. Let X be a random variable with distribution function F. Then $|X|$ is a random variable. Find its distribution function.
For $y < 0$, $\mathbb{P}(|X| \leq y) = 0$. For $y \geq 0$,

$$\mathbb{P}(|X| \leq y) = \mathbb{P}\{-y \leq X \leq y\} = \mathbb{P}(X \leq y) - \mathbb{P}(X < -y) = F(y) - F(-y) + \mathbb{P}\{X = -y\}$$

Let $X^+ = \begin{cases} X, & X \geq 0 \\ 0, & X < 0 \end{cases}$ and $X^- = \begin{cases} 0, & X > 0 \\ X, & X \leq 0. \end{cases}$ Then

$$\mathbb{P}(X^+ \leq y) = \begin{cases} 0 & \text{if } y < 0 \\ \mathbb{P}(X \leq 0) & \text{if } y = 0 \\ \mathbb{P}(X < 0) + \mathbb{P}\{0 \leq X \leq y\} & \text{if } y > 0, \end{cases}$$

and

$$\mathbb{P}\{X^- \leq y\} = \begin{cases} 1 & \text{if } y \geq 0, \\ \mathbb{P}\{X \leq y\} & \text{if } y < 0. \end{cases}$$

Example 6.2. Let X be a random variable with probability distribution function $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \ -\infty < x < \infty$. Let $Y = X^2$. In this case $g'(x) = 2x$ which is positive for $x > 0$ and negative for $x < 0$, so that the conditions of Theorem 5.1 are not satisfied. But for $y > 0$,

$$\mathbb{P}(Y \leq y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = F(\sqrt{y}) - F(-\sqrt{y}).$$

Thus the pdf of Y is given by,

$$h(y) = \begin{cases} \frac{1}{2\sqrt{y}}(F(\sqrt{y}) - F(-\sqrt{y})), & y > 0 \\ 0, & y \leq 0 \end{cases} = \begin{cases} \frac{1}{\sqrt{2\pi y}} e^{-\frac{y}{2}}, & y > 0 \\ 0, & y \leq 0. \end{cases}$$

Example 6.3. Let X be a random variable with uniform distribution on $[-1, 1]$, that is, the pdf of X is $f(x) = \frac{1}{2}, \ -1 \leq x \leq 1$, and $= 0$ *otherwise*. Let $Y = X^+$. Then from example 5.2,

$$\mathbb{P}(\{Y \leq y\}) = \begin{cases} 0, & y < 0 \\ \frac{1}{2}, & y = 0 \\ \frac{1}{2} + \frac{1}{2}y, & 1 \geq y > 0 \\ 1, & y > 1. \end{cases}$$

We see that the distribution function of Y has a jump at $y = 0$ and that Y is neither discrete nor continuous. Note that, all we require is that $\mathbb{P}(X < 0) > 0$ for X^+ to be of mixed type.

Theorem 6.3. Let X be a random variable of the continuous type with pdf f. Let $y = g(x)$ be differentiable and strictly monotone for all x. Then $Y = g(X)$ is also a random variable of the continuous type with pdf given by

$$h(y) = \begin{cases} f[g^{-1}(y)] \left| \frac{d}{dy} g^{-1}(y) \right|, & \alpha < y < \beta \\ 0, & otherwise, \end{cases}$$

where $\alpha = min\{g(-\infty), g(+\infty)\}$ and $\beta = max\{g(-\infty), g(+\infty)\}$.
Proof: If g is differentiable and strictly increasing for all x, then the limits α, β exist (may be infinite) and the inverse function $x = g^{-1}(y)$ exists, is strictly increasing and differentiable. The distribution function of Y for $\alpha < y < \beta$ is given by

$$\mathbb{P}(Y \leq y) = \mathbb{P}(X \leq g^{-1}(y)).$$

The pdf of Y is given by

$$h(y) = \frac{d}{dy}\mathbb{P}(Y \leq y) = f[g^{-1}(y)]\frac{d}{dy}(g^{-1}(y)).$$

Similarly, if g is strictly decreasing, we have

$$\mathbb{P}(Y \leq y) = \mathbb{P}(X \geq g^{-1}(y)) = 1 - \mathbb{P}(X \leq g^{-1}(y)) \qquad [\because X \text{ is continuous random variable}],$$

so that

$$h(y) = -\frac{d}{dy}\mathbb{P}(Y \leq y) = f[g^{-1}(y)]\frac{d}{dy}(g^{-1}(y)).$$

Since, g and g' are both strictly decreasing, $\frac{d}{dy}[g^{-1}(y)]$ is negative and hence the result follows.

Theorem 6.4. Let X be a random variable of the continuous type with pdf f. Let $y = g(x)$ be differentiable for all x and assume that $g'(x)$ is continuous and non-zero at all but a finite number of values of x. If for every real number y,

(a) there exist a +ve integer $n = n(y)$ and real numbers $x_1(y), x_2(y), \ldots, x_n(y)$ such that

$$g(x_k(y)) = y, \ g'(x_k(y)) \neq 0, \ k = 1, 2, 3, \ldots, n(y),$$

or

(b) there does not exist any x such that $g(x) = y,\ g'(x) \neq 0$, in which case we write $n(y) = 0$.

Then Y is a continuous random variable with pdf given by

$$h(y) = \begin{cases} \sum_{k=1}^{n} f(x_k(y))\,|g'(x_k(y))|^{-1}, & if\ n > 0 \\ 0, & otherwise. \end{cases}$$

Example 6.4. Let X be a random variable with pdf f, and let $Y = |X|$.
Here, $n = 2, X_1(y) = y, x_2(y) = y$ for $y > 0$, and

$$h(y) = \begin{cases} f(y) + f(-y), & y > 0 \\ 0, & y \leq 0. \end{cases}$$

In particular, if $f(x) = \frac{-1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}},\ -\infty < x < 0$, then $h(y) = \begin{cases} \frac{2}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}, & y > 0 \\ 0, & otherwise. \end{cases}$

Example 6.5. Let X be a random variable with pdf $f(x) = \begin{cases} \theta e^{-\theta x}, & x > 0 \\ 0, & x < 0, \end{cases}$ where $\theta > 0$. Let $Y = sin(X)$ and let $sin^{-1}(y)$ be the principal values. Then for $0 < y < 1$,

$$\begin{aligned}
\mathbb{P}(Y \leq y) &= \mathbb{P}(sin(X) \leq y) \\
&= \mathbb{P}(0 < X \leq sin^{-1}(y) \text{ or } (2n-1)\pi - sin^{-1}(y) \leq X \leq 2n\pi + sin^{-1}(y) \text{ for all integers } n \geq \\
&= \mathbb{P}(0 \leq X \leq sin^{-1}(y)) + \sum_{n=1}^{\infty} \mathbb{P}((2n-1)\pi - sin^{-1}(y) \leq X \leq 2n\pi + sin^{-1}(y)) \\
&= 1 - e^{-\theta sin^{-1}(y)} + \sum_{n=1}^{\infty} [e^{-\theta[(2n-1)\pi - sin^{-1}y]} - e^{-\theta[2n\pi + sin^{-1}(y)]}] \\
&= 1 - e^{-\theta sin^{-1}(y)} + (e^{\theta\pi + \theta sin^{-1}(y)} - e^{-\theta sin^{-1}(y)}) \sum_{n=1}^{\infty} e^{-2\theta\pi n} \\
&= 1 - e^{-\theta sin^{-1}(y)} + (e^{\theta\pi + \theta sin^{-1}(y)} - e^{-\theta sin^{-1}(y)}) \frac{e^{-2\theta\pi}}{1 - e^{-2\theta\pi}} \\
&= 1 + \frac{e^{-\theta\pi + \theta sin^{-1}y - e^{-\theta sin^{-1}y}}}{1 - e^{-2\pi\theta}}.
\end{aligned}$$

Taking the derivative w.r.t y gives the pdf of y as

$$h(y) = \begin{cases} \theta e^{-\theta\pi}(1 - e^{-2\theta\pi})^{-1}(1 - y^2)^{-\frac{1}{2}}[e^{\theta sin^{-1}y} + e^{-\theta\pi - \theta sin^{-1}y}] & if\ -1 < y < 0 \\ \theta(1 - e^{-2\theta\pi})(1 - y^2)^{-\frac{1}{2}}[e^{-\theta sin^{-1}y + e^{-\theta\pi + \theta sin^{-1}y}}] & if\ 0 < y < 1 \\ 0 & otherwise. \end{cases}$$

Theorem 6.5. Let $(X_1, X_2, \ldots, X_n)$ be an $n - dimensional$ random vector with pdf $f(x_1, x_2, \ldots, x_n)$

(a) Let

$$\begin{aligned} y_1 &= g_1(x_1, x_2, \ldots, x_n), \\ y_2 &= g_2(x_1, x_2, \ldots, x_n), \\ \vdots &= \vdots \qquad \vdots \qquad \vdots \\ y_n &= g_n(x_1, x_2 \ldots, x_n) \end{aligned}$$

be a one-to-one mapping of $\mathbb{R}^n$ into itself, i.e., there exists the inverse transformation

(b) Let

$$\begin{aligned} x_1 &= h_1(y_1, y_2, \ldots, y_n), \\ x_2 &= h_2(y_1, y_2, \ldots, y_n), \\ \vdots &= \vdots \qquad \vdots \qquad \vdots \\ x_n &= g_n(y_1, y_2 \ldots, y_n) \end{aligned}$$

defined over the range of the transformation,

(c) Assume that both the mapping and its inverse are continuous,

(d) Assume that the Jacobian J of the inverse transformation

$$J = \frac{\partial(x_1, x_2, \ldots, x_n)}{\partial(y_1, y_2, \ldots, y_n)} = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \cdots & \frac{\partial x_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \cdots & \frac{\partial x_n}{\partial y_n} \end{vmatrix}$$

is different from 0 for $(y_1, y_2, \ldots, y_n)$ in the range of the transformation.

Then $(Y_1, Y_2, \ldots, Y_n)$ has a joint absolutely continuous distribution function with pdf given by

$$w(y_1, y_2, \ldots, y_n) = |J| f(h_1(y_1, y_2, \ldots, y_n), \ldots, h_n(y_1, y_2, \ldots, y_n)).$$

Example 6.6. Let X_1, X_2 be independent random variables with common density given by

$$f(x) = \begin{cases} 1 & if\ 0 < x < 1 \\ 0 & otherwise. \end{cases}$$

FInd the joint distribution of $X_1 + X_2$.
Let $Y_1 = X_1 + X_2, Y_2 = X_1 - X_2$, then the jacoobian of the transformation is given by

$$J = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = -\frac{1}{2},$$

and the joint density of Y_1, Y_2 is given by

$$\begin{aligned} f_{Y_1,Y_2}(y_1,y_2) &= \frac{1}{2} f\left(\frac{y_1+y_2}{2}\right) f\left(\frac{y_1-y_2}{2}\right) \;\; if \; 0 < \frac{y_1+y_2}{2} < 1 \; or \; 0 < \frac{y_1-y_2}{2} < 1 \\ &= \frac{1}{2} \;\; if \; (y_1,y_2) \in \{(y_1,y_2) \in \mathbb{R}^2 : 0 < y_1+y_2 < 2 \; or \; 0 < y_1-y_2 < 2\}. \end{aligned}$$

The marginal pdf of Y_1 and Y_2 are given by

$$f_{Y_1}(y_1) = \begin{cases} \int_{-y_1}^{y_1} \frac{1}{2} dy_2 = y_1, & 0 < y_1 \leq 1 \\ \int_{y_1-2}^{2-y_1} \frac{1}{2} dy_2 = 2 - y_1, & 1 < y_1 < 2 \\ 0, & otherwise, \end{cases}$$

$$f_{Y_2}(y_2) = \begin{cases} \int_{-y_2}^{y_2+2} \frac{1}{2} dy_1 = y_2 + 1, & -1 < y_2 \leq 0 \\ \int_{y_2}^{2-y_2} \frac{1}{2} dy_1 = 1 - y_2, & 0 < y_2 < 1 \\ 0, & otherwise. \end{cases}$$

Exercise: Let X_1, X_2 be independent random variables with common pdf as standard normal $N(0,1)$. Identify the distribution function of $X_1 - X_2$ and $X_1 + X_2$.

Exercise: X is a normal variate with mean 1 and variance 4, Y is another normal variate independent of X with mean 2 and variance 3. What is the distribution of $X + 2Y$?
Ans: $X + 2Y \overset{D}{\sim} N(5, 16)$.

7 Correlation, Moment, Skewness, and Kurtosis

Recall that $\mathbb{E}[(Y-a)^2]$ is minimized when we choose $a = \mathbb{E}(Y)$, so that $\mathbb{E}[Y]$ may be interpreted as best constant predictor of Y. If instead, we choose to predict Y by a linear function of X, say $aX + b$ and measure the error of this prediction as $\mathbb{E}[(Y - aX - b)^2]$, then we should choose a and b to minimize this so called **mean square error**. Clearly, $\mathbb{E}[(Y - aX - b)^2]$ is minimized for any a, by changing $b = \mathbb{E}[(Y - aX)] = \mathbb{E}[Y] - a\mathbb{E}[X]$.
With this choice of b, we find a such that

$$\begin{aligned}\mathbb{E}[(Y - aX - b)^2] &= \mathbb{E}[(Y - aX - \mathbb{E}[Y] + a\mathbb{E}[X])^2] = \mathbb{E}[(Y - \mathbb{E}(Y) - a(X - \mathbb{E}(X)))^2] \\ &= \sigma_Y^2 - 2a\mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] + a^2\sigma_X^2 = \sigma_Y^2 - 2am_{11} + a^2\sigma_X^2\end{aligned}$$

is minimum.
An easy computation shows that the minimum occurs if we choose $a = \frac{m_{11}}{\sigma_X^2}$, provided $\sigma_X^2 > 0$. Moreover,

$$\min_{a,b} \mathbb{E}[(Y - aX - b)^2] = \min_a \{\sigma_Y^2 - 2am_{11} + a^2\sigma_X^2\} = \frac{\sigma_X^2\sigma_Y^2 - m_{11}^2}{\sigma_X^2} = \sigma_Y^2\left(1 - \left(\frac{m_{11}}{\sigma_X\sigma_Y}\right)\right).$$

Let us write $\rho = \frac{m_{11}}{\sigma_X\sigma_Y}$. Then the above equation shows that predicting Y by a linear function of X reduces the prediction error from σ_Y^2 to $\sigma_Y^2(1-\rho^2)$. We, therefore, think of ρ as the measure of linear departure between the random variables X and Y.

Definition 7.1 (Correlation Coefficient)**.** The **correlation coefficient** between X and$Y$ is defined as

$$\rho = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\sigma_X\sigma_Y}.$$

Theorem 7.1. :

(a) The correlation coefficient ρ between random variables X and Y satisfies $|\rho| \leq 1$.

(b) The equality $|\rho| = 1$ holds iff there exists constants $a \neq 0$ and b such that $\mathbb{P}(aX + b = Y) = 1$.

Proof: (a) Using Cauchy-Schwarz inequality on $Cov(X,Y)^2$, we get

$$\begin{aligned}|\rho|^2 &= \left|\frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\sigma_X\sigma_Y}\right|^2 = \left|\frac{Cov(X,Y)^2}{Var(X)Var(Y)}\right| = \frac{|\mathbb{E}[(X - [E](X))(Y - \mathbb{E}(Y))]^2|}{Var(X)Var(Y)} \\ &\leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]\mathbb{E}[(Y - \mathbb{E}[Y])^2]}{Var(X)Var(Y)} = \frac{Var(X)Var(Y)}{Var(X)Var(Y)} = 1.\end{aligned}$$

And hence, $|\rho| \leq 1$.
(b) The equality holds iff an equality holds for the Cauchy-Schwarz inequality iff there exists $a, b \in \mathbb{R}$ such that $aX + b = Y$.

Theorem 7.2. Let $\mathbb{E}[X^2] < \infty, \mathbb{E}[Y^2] < \infty$, and let $U = aX + b, V = cY + d$. Then

$$\rho_{X,Y} = \pm\rho_{U,V},$$

where $\rho_{X,Y}$ and $\rho_{U,V}$ are the respective correlation coefficients.
Proof: The proof is trivial by applying linearity of expectation and variance on $\rho_{U,V}$.

Example 7.1. Let (X, Y) be jointly distributed with density function

$$f(x) = \begin{cases} x + y, & 0 < x < 1, 0 < y < 1 \\ 0, & otherwise. \end{cases}$$

Find the correlation coefficient between the random variables X and Y.

Definition 7.2 ($k^{th}-$ moment)**.** Let k be a primitive integer and c be a constant. If $\mathbb{E}(X - c)^k$ exists, we call it the **moment** of order k at point c.

Theorem 7.3. If the moment of order t exists for a random variable X, the moments of order $0 < s < t$ exist.
Proof: Let X be of continuous type with pdf f, we have

$$\mathbb{E}[|X|^s] = \int_{|x|^s \leq 1} |x|^s f(x)dx + \int_{|x|^s > 1} |x|^s f(x)dx \leq \mathbb{P}(|X|^s \leq 1) + \mathbb{E}[|X|^t] < \infty.$$

A similar argument also holds for discrete random variable.

Example 7.2. Let X be a Cauchy random variable with pdf

$$f(x) = \frac{1}{\mu}\frac{1}{1 + x^2}, \quad -\infty < x < \infty.$$

Then $\mathbb{E}[X^k]$ does not exists for any $k > 1$.
By above theorem, it is sufficient to prove that $\mathbb{E}[X]$ does not exist.

$$\mathbb{E}[X] = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x}{1 + x^2}dx = \lim_{\substack{a \to -\infty, \\ b \to \infty}} \frac{1}{\pi}\int_a^b \frac{x}{1 + x^2}dx = \lim_{\substack{a \to -\infty, \\ b \to \infty}} \left(\frac{b - a}{2\pi}\right).$$

Thus, we can clearly see that $\mathbb{E}[X]$ does not exists.

Definition 7.3 (Central Moment)**.** If we take $c = \mathbb{E}[X] = \mu$, which exists since $\mathbb{E}[X] < \infty$, we call $\mathbb{E}[(X - \mu)^k]$ the **central moment** of order k or the moment of order k about the mean. We shall write it as $\sigma_k = \mathbb{E}[(X - \mu)^k]$.

A general expression of the kth moment and central moment.

$$\begin{aligned}
\sigma_k &= \mathbb{E}[(X - m_1)^k] = \mathbb{E}\left[\binom{k}{0}X^k - \binom{k}{1}X^{k-1}m_1 + \binom{k}{2}X^{k-2}m_1^2 - \ldots + (-1)^k\binom{k}{k}m_1^k\right] \\
&= m_k - \binom{k}{1}m_{k-1}m_1 + \binom{k}{2}m_{k-2}m_1^2 - \ldots + (-1)^k m_1^k, \\
m_k &= \mathbb{E}[X^k] = \mathbb{E}[(X - \mu + \mu)^k] \\
&= \mathbb{E}\left[\binom{k}{0}(X - \mu)^k + \binom{k}{1}(X - \mu)^{k-1}\mu + \binom{k}{2}(X - \mu)^{k-2}\mu^2 + \ldots + \binom{k}{k}\mu^k\right] \\
&= \sigma_k + \binom{k}{1}\sigma_{k-1}\mu + \binom{k}{2}\sigma_{k-2}\mu^2 + \ldots + \mu^k.
\end{aligned}$$

Definition 7.4 (Product Moment)**.** If $\mathbb{E}[X^jY^k] < \infty$ for non-negative integers j and k, we call $\mathbb{E}[X^jY^k]$ the **product moment** of order $(j+k)$ of (X,Y) and write

$$m_{jk} = \mathbb{E}[X^jY^k],$$

$$m_{10} = \mathbb{E}[X], m_{01} = \mathbb{E}[Y], m_{20} = \mathbb{E}[X^2], m_{11} = \mathbb{E}[XY], m_{02} = \mathbb{E}[Y^2].$$

Definition 7.5 (Central Product Moment)**.** If $\mathbb{E}[(X-\mathbb{E}[X])^j(Y-\mathbb{E}[Y])^k] < \infty$ for non-negative integers j and k, we call $\mathbb{E}[(X-\mathbb{E}[X])^j(Y-\mathbb{E}[Y])^k]$ the **product moment** of order $(j+k)$ of (X,Y) and write

$$\sigma_{jk} = \mathbb{E}[(X-\mathbb{E}[X])^j(Y-\mathbb{E}[Y])^k],$$

$$\sigma_{10} = \sigma_{01} = 0, \sigma_{20} = Var(X), \sigma_{02} = Var(Y), \sigma_{11} = Cov(X,Y).$$

Theorem 7.4. $Var(X) < \mathbb{E}[(X-c)^2]$ for any $c \neq \mathbb{E}[X]$.
Proof: For any $c \neq \mathbb{E}[X]$,

$$\begin{aligned} Var(X) &= \mathbb{E}[(X-\mu)^2] = \mathbb{E}[(X-c)^2 + (c-\mu)^2 + 2(X-c)(c-\mu)] \\ &= \mathbb{E}[(X-c)^2] - (c-\mu)^2 < \mathbb{E}[(X-c)^2]. \end{aligned}$$

Definition 7.6. Let X be a random variable with distribution function F. A number x satisfying

$$\frac{1}{2} \leq F(x) \leq \frac{1}{2} + \mathbb{P}(X = x)$$

or, equivalently,

$$\mathbb{P}(X \leq x) \geq \frac{1}{2} \text{ and } \mathbb{P}(X \geq x) \geq \frac{1}{2}$$

is called **median** of X (or F). A median is not necessarily unique.

Example 7.3. Let $\mathbb{P}(X=2) = \mathbb{P}(X=0) = \frac{1}{4}$, $\mathbb{P}(X=1) = \frac{1}{3}, \mathbb{P}(X=2) = \frac{1}{6}$. Then $\mathbb{P}(X \leq 0) = \frac{1}{2}$ and $\mathbb{P}(X \geq 0) = \frac{3}{4} \neq \frac{1}{2}$. But for any $0 < x < 1$, $\mathbb{P}(X \leq x) = \frac{1}{2}$ and $\mathbb{P}(X \geq x) = \frac{1}{2}$. Thus, every x in interval $(0,1)$ is a median.

Definition 7.7 (Factorial Moment)**.** $\mathbb{E}[X(X-1)\cdots(X-k+1)]$, for various values of k, is the $k^{th}-$ **factorial moment** of the random variable X.

Definition 7.8. A random variable X satisfying

$$\mathbb{P}(X \leq x) \geq p, \mathbb{P}(X \geq x) \geq 1-p, \ 0 < p < 1,$$

is called a **quantile** of order p[or $(100p)^{th}$ percentile] for the random variable X (or the distribution function of X). We write ${}_p(X)$ for a quantile of order p for a random variable X. If x is a quantile of order p for a random variable X with distribution function F, then

$$p \leq F(x) \leq p + \mathbb{P}(X = x).$$

If X is of continuous type the last equation becomes $F(x) = p$. If F is strictly increasing for a continuous random variable, then $F(x) = p$ has a unique solution. Otherwise it may have any number (even uncountable) of solutions as quantile.

Definition 7.9 (Skewness)**.** Let X be a random variable with $k^{th}-$ moment $m_k = \mathbb{E}[X^k]$ and variance σ^2, then the **skewness** of X is defined by

$$skew(X) = \frac{\mathbb{E}[(X - m_1)^3]}{\sigma^3}$$

. It tells about the asymmetry of a distribution function. If the skewness of a random variable is negative, its distribution function is **left-skewed**; if the skewness is zero, the distribution function is symmetrical around its mean; and if the skewness is positive, its distribution function is **right-skewed**.

Theorem 7.5. Let X be a random variable with $k^{th}-$ moment $m_k = \mathbb{E}[X^k]$ and variance σ^2, then

(a) $skew(X) = \frac{m_3 - 3m_1m_2 + 2m_1^3}{\sigma^3}$,

(b) $skew(S_n) = \frac{X_1}{\sqrt{n}}$ where $S_n = \sum_{i=1}^n X_i$ is sum of $i.i.d$ random variables,

(c) $skew(X) = \frac{(1-2p)}{\sqrt{npq}}$, when $X \overset{D}{\sim} Bin(n, p)$,, where $p + q = 1$,

(d) $skew(X) = \frac{1}{\sqrt{\lambda}}$, when $X \overset{D}{\sim} Poisson(\lambda)$,

(e) $skew(X) = \frac{2}{\sqrt{t}}$, when $X \overset{D}{\sim} \Gamma(\lambda, t)$ and t is integral.

Definition 7.10 (Kurtosis)**.** Let X be a random variable with $k^{th}-$ moment $m_k = \mathbb{E}[X^k]$ and variance σ^2, then the **kurtosis** of X is defined by

$$kur(X) = \frac{\mathbb{E}[(X - m_1)^4]}{\sigma^4}.$$

Kurtosis is a statistical measure which defines how heavily the tails of a distribution differs from those of normal distribution.

Theorem 7.6. (a) $kur(X) = 3$, when $X \overset{D}{\sim} N(\mu, \sigma^2)$,

(b) $kur(X) = 9$, when $X \overset{D}{\sim} exp(\lambda)$,

(c) $kur(X) = 3 + \frac{1}{\lambda}$, when $X \overset{D}{\sim} Poisson(\lambda)$,

(d) $kur(S_n) = \frac{3}{n} + \frac{kur(X_1) - 3}{n}$, where $S_n = \sum_{i=1}^n X_i$ is sum of $i.i.d$ random variables.

Definition 7.11 (Symmetric Random Variable)**.** We say that a random variable X is **symmetric** about a point α if

$$\mathbb{P}(X \geq \alpha + x) = \mathbb{P}(X \leq \alpha - x) \ \forall \ x \in \mathbb{R}.$$

In terms of distribution function of X, this means that if

$$F(\alpha - x) = 1 - F(\alpha + x) + \mathbb{P}(X = \alpha + x) \ \forall \ x \in \mathbb{R}.$$

We say that the random variable X is symmetric with α as the **center of symmetry**. In particular, if X is a random variable of continuous type, X is symmetric with center α if and only if the pdf of X satisfies

$$f(\alpha - x) = f(\alpha + x) \ \forall \ x \in \mathbb{R}.$$

If $\alpha = 0$, we simply say that X is symmetric (or that F is symmetric).

8 Probability Generating Functions

Definition 8.1 (Probability Generating Function)**.** The **probability generating function** of the random variable X is defined to be the generating function

$$G(z) = E(z^X)$$

of its probability mass function.

Note that, G does indeed generate the sequence $\{fX(i) : i \geq 0\}$ since

$$E(z^X) = \sum_i z^i P(X = i) = \sum_i z^i f_X(i).$$

We write G_X when we wish to stress the role of x. If X takes values in the non-negative integers, its generating function G_X converges at least when $|z| \leq 1$ and sometimes in a larger interval. Generating functions can be defined for random variables taking negative as well as positive integer values. Such generating functions generally converges for values of z satisfying $\alpha < |z| < \beta$ for some α, β such that $\alpha \leq 1 \leq \beta$. We shall make occasional use of such generating functions, but we do not develop their theory systematically.
In advance of giving examples and applications of the method of generating functions, we recall some basic properties of power series. Let $G(z) = \sum_{i=0} a_i z^i$ where $a = \{a_i : i \geq 0\}$ is a real sequence.
Convergence: There exist a radius of convergence $R(\geq 0)$ such that the sum converges absolutely if $|z| < R$ and diverges if $|z| > R$. The sum is uniformly convergent on sets of the form $\{z : |z| \leq R\}$ for any $R' < R$.
Differentiation: $G_a(z)$ may be differentiated or integrated term by term any number of times at points at z satisfying $|z| < R$.
Uniquenss: If $G_a(z) = G_b(z)$ for $|z| < R'$ where $0 < R' \leq R$ then $a_n = b_n \ \forall \ n$. Furthermore,

$$a_n = \frac{1}{n!} G_a^{(n)}(0).$$

Abel's Theorem: If $a_i \geq 0 \forall \ i$ and $G_a(z)$ is finite for $|z| \leq 1$, then

$$\lim_{z \uparrow 1} G_a(z) = \sum_{i=0}^{\infty} a_i,$$

whether the sum is finite or equal to ∞. This standard result is useful when the radius of convergence R satisfied $R = 1$, since the one has no a priori right to take the limit as $s \uparrow 1$. In particular, the radius of convergence of a probability generating function is at least 1.

Example 8.1. Probability generating functions of some random variable:
Bernoulli variable:

$$G(z) = 1 - p + pz = pz + q \text{ where } q = 1 - p,$$

Binomial variable:

$$G(z) = \sum_{i=0}^{n} z^i \binom{n}{i} p^i (1-p)^{n-i} = (1-p+pz)^n = (q+pz)^n, \text{ where } q = 1-p,$$

Poisson variable:

$$G(z) = E(z^X) = \sum_{k=0}^{\infty} \frac{z^k e^{-\lambda} \lambda^k}{k!} = e^{\lambda(z-1)},$$

Geometric variable:

$$G(z) = E(z^X) = \sum_{k=1}^{\infty} z^k p(1-p)^{k-1} = \frac{pz}{1-z(1-p)} = \frac{pz}{1-qz} \text{ where } q = 1-p.$$

Theorem 8.1. If X has a generating function $G(z)$, then
(a) $E(X) = G'(1)$
(b) More generally, $E[X(X-1)\dots(X-k+1)] = G^{(k)}(1)$
(c) $Var(X) = G''(1) + G'(1) - [G'(1)]^2$.
Proof: The proof is trivial.

Theorem 8.2. If X and Y are independent random variables, then $G_{X+Y}(z) = G_X(z)G_Y(z)$. More generally, if $X_1, X_2, \dots, X_n$ are independent random variables, then $G_{X_1+X_2+\dots+X_n} = G_{X_1}(z)G_{X_2}(z)\dots G_{X_n}(z)$.

Proof:

$$\begin{aligned} G_{X+Y}(z) = E(z^{X+Y}) &= \sum_x \sum_y z^x z^y f_{X,Y}(x,y) \\ &= \sum_x z^x f_X(x) \sum_y z^y f_Y(y) \quad [\because f_{X,Y}(x,y) = f_X(x)f_Y(y)] \\ &= G_X(z)G_Y(z) \end{aligned}$$

We can proof the case for n random variables using induction.
The converse of this statement may not be true.

Example 8.2. If X and Y are independent random variables following the Poisson distributions with parameters λ and μ, respectively. The distribution of $X+Y$ can be found using probability generating function as follows:

$$G_{X+Y}(z) = G_X(z)G_Y(z) = e^{\lambda(z-1)} \cdot e^{\mu(z-1)} = e^{(\lambda+\mu)(z-1)}.$$

This gives, $X + Y \overset{D}{\sim} Poisson(\lambda + \mu)$.

Theorem 8.3. If $X_1, X_2, \dots$ is a sequence of independent identically distributed random variables with common p.g.f. $G_X(z)$ and $N(\geq 0)$ is a random variable which is independent of X_i and had p.g.f. $G_N(z)$, then $S = X_1 + X_2 + \dots + X_N$ has the p.g.f. $G_s(z) = G_N(G_X(z))$.

Proof: Rearranging the terms of p.g.f of S, we get

$$\begin{aligned}
G_S(z) &= E(z^S) = E[z^{X_1+X_2+\cdots+X_N}] \\
&= E_N[E_{X|N}(z^{X_1+X_2+\cdots+X_N}|N)] \\
&= \sum_n E(z^{X_1+X_2+\cdots+X_N}|N=n)P(N=n) \\
&= \sum_n [G_X(z)]^n P(N=n) = G_N(G_X(z)).
\end{aligned}$$

Example 8.3. A hen lays N eggs, where N is Poisson distributed with parameter λ. Each egg hatches with probability p, independently of all other eggs. Let k be the number of chicks. Then $K = X_1 + X_2 + \cdots + X_N$ where $X_1, X_2, \ldots, X_N$ are independent Bernoulli variables with parameter p. The distribution of the random variable K can be found to be

$$G_N(z) = e^{\lambda(z-1)}$$

$$G_K(z) = G_N(G_X(z)) = G_N(q+pz) = e^{\lambda p(z-1)} \text{ where } q = 1-p.$$

Thus, $K \overset{D}{\sim} Poisson(\lambda p)$.

Definition 8.2 (Joint Probability Generating Function)**.** The **joint probability generating function** of the random variables X_1 and X_2 taking values in the non-negative integers is defined by

$$G_{X_1,X_2}(z_1, z_2) = E(z_1^{X_1} \cdot z_2^{X_2}).$$

This definition can be extended for an arbitrary family of random variables.

Theorem 8.4. Random variables $X1$ and $X2$ are independent if and only if

$$G_{X_1,X_2}(z_1, z_2) = G_{X_1}(z_1) \cdot G_{X_2}(z_2).$$

Proof: Proof is simple in the forward direction and is given as follows:

$$\begin{aligned}
G_{X_1,X_2}(z_1, z_2) &= E(z_1^{X_1} \cdot z_2^{X_2}) = \sum_i \sum_j z_1^i z_2^j f_{X_1,X_2}(i,j) \\
&= \sum_i z_1^i f_{X_1}(i) \sum_j z_2^j f_{X_2}(j) \quad [\because f_{X_1,X_2}(i,j) = f_{X_1}(i) f_{X_2}(j)] \\
&= G_{X_1}(z_1) G_{X_2}(z_2).
\end{aligned}$$

To prove the converse, equate the coefficients of the terms such as $z_1^i z_2^j$ to deduce after some manipulation that $f_{X_1,X_2}(i,j) = f_{X_1}(i) f_{X_2}(j)$.

9 Moment Generating Functions

Probability generating functions proved to be a very useful in handling non-negative integral random variables. For more general variables X, it is natural to make the substitution $z = e^t$ in the quantity $G_X(z) = E(z^X)$.

Definition 9.1 (Moment Generating Function)**.** The **moment generating function** of a variable X is the function $M : \mathbb{R} \to [0, \infty)$ given by

$$M(t) = E(e^{tX}).$$

Moment generating functions are related to Laplace transforms since $M(t) = \int e^{tX} dF_X(x) = \int e^{tX} f_X(x)dx$ if X is continuous with density functions f_X. M.g.f. have properties similar to those of probability generating functions.

Theorem 9.1. If $M(t) < \infty$ on some open interval containing the origin, then

(a) $E(X) = M^{(1)}(0), E(X^k) = M^{(k)}(0)$.

(b) The function M can be expanded via Taylor's series within its circle of convergence

$$M(t) = \sum_{k=0}^{\infty} \frac{E(X^k)}{k!} t^k,$$

which is to say that M is the 'exponential generating function' of the sequence of moments of X.

(c) If X and Y are independent, then $M_{X+Y}(t) = M_X(t)M_Y(t)$ because

$$\begin{aligned} M_{X+Y}(t) = E(e^{t(X+Y)}) &= \int\int e^{t(x+y)} f_{X,Y}(x,y)dxdy \\ &= \int e^{tx} f_X(x)dx \int e^{ty} f_Y(y)dy = M_X(t)M_Y(t). \end{aligned}$$

The converse of this statement is not usually true.

(d) More generally, if $X_1, X_2, \ldots, X_n$ are independent random variables, then

$$M_{X_1+X_2+\cdots+X_n}(t) = M_{X_1}(t) \cdot M_{X_2}(t) \cdots M_{X_n}(t).$$

Proof: The proof is trivial.

Example 9.1. 1. If $X \overset{D}{\sim} \exp(\lambda)$, then m.g.f. of X can be obtained as

$$M_X(t) = \lambda \int_0^{\infty} e^{tx} e^{-\lambda x} dx = \frac{\lambda}{t-\lambda} e^{-\lambda x + tx}|_0^{\infty} = \frac{\lambda}{\lambda - t}, \ t < \lambda,$$

2. If $X \overset{D}{\sim} N(\mu, \sigma^2)$, then m.g.f. of X can be defined as

$$
\begin{aligned}
M_X(t) &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}(x^2-2\mu x+\mu^2-2\sigma^2 tx)} dx \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}(x^2-2x(\mu+\sigma^2 t)+\mu^2+2\mu\sigma^2 t-2\mu\sigma^2 t+\sigma^4 t^2-\sigma^4 t^2)} dx \\
&= \frac{e^{\mu t+\frac{\sigma^2 t^2}{2}}}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}(x-\mu-\sigma^2 t)^2} dx \\
&= \frac{e^{\mu t+\frac{\sigma^2 t^2}{2}}}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{v^2}{2}} \sigma dv \qquad\qquad [Put\ \frac{x-\mu-\sigma^2 t}{\sigma} = v] \\
&= e^{\mu t+\frac{\sigma^2 t^2}{2}}, \text{ where } -\infty < t < \infty.
\end{aligned}
$$

Definition 9.2 (Joint Moment Generating Function)**.** The **joint moment generating function** of random variable X and Y is the function $M_{X,Y} : \mathbb{R}^2 \to [0, \infty)$ given by

$$M_{X,Y}(s,t) = E(e^{sx+ty}) = \int\int e^{sx+ty} f_{X,Y}(x,y)dxdy.$$

Let $X_1, X_2, \ldots, X_n$ be jointly distributed. If $E[e^{\sum_{j=1}^n t_j x_j}]$ exists for $|t_j| \leq h_j, j = 1, 2, \ldots, n$ for some $h_j > 0, j = 1, 2, \ldots, n$, we write

$$M_{X_1,X_2,\ldots,X_n}(t_1, t_2, \ldots, t_n) = E(e^{t_1X_1+t_2X_2+\cdots+t_nX_n})$$

and call it the moment generating function of $(X_1, X_2, \ldots, X_n)$.

Theorem 9.2. The joint moment generating function $M_{X_1,X_2}(t_1, t_2)$ uniquely determines the joint distribution of (X_1, X_2) and conversely if the m.g.f. exists, it is unique.

Corollary: The joint m.g.f. $M_{X_1,X_2}(t_1, t_2)$ completely determines the marginal distributions of X and Y. Indeed

$$
\begin{aligned}
M_{X_1,X_2}(t_1, 0) &= E(e^{t_1X_1}) = M_{X_1}(t_1) \\
M_{X_1,X_2}(0, t_2) &= E(e^{t_2X_2}) = M_{X_2}(t_2).
\end{aligned}
$$

Theorem 9.3. If $M_{X_1,X_2}(t_1, t_2)$ exists, the moments of all orders of (X, Y) exist and may be obtained from

$$\frac{\partial^{m+n}}{\partial t_1^m \partial t_2^n} \left(M_{X_1,X_2}(t_1, t_2)\right)\Big|_{t_1=t_2=0} = E(X_1^m X_2^n).$$

Thus,

$$\begin{aligned}
\frac{\partial}{\partial t_1} M_{X_1,X_2}(t_1,t_2)\Big|_{t_1=t_2=0} &= E(X_1),\\
\frac{\partial}{\partial t_2} M_{X_1,X_2}(t_1,t_2)\Big|_{t_1=t_2=0} &= E(X_2),\\
\frac{\partial^2}{\partial t_1^2} M_{X_1,X_2}(t_1,t_2)\Big|_{t_1=t_2=0} &= E(X_1^2),\\
\frac{\partial^2}{\partial t_2^2} M_{X_1,X_2}(t_1,t_2)\Big|_{t_1=t_2=0} &= E(X_2^2),\\
\frac{\partial^2}{\partial t_1 \partial t_2} M_{X_1,X_2}(t_1,t_2)\Big|_{t_1=t_2=0} &= E(X_1X_2).
\end{aligned}$$

Theorem 9.4. The random variables X_1 and X_2 are independent if and only if $M_{X_1,X_2}(t_1,t_2) = M_{X_1}(t_1)M_{X_2}(t_2)$.
Proof: If X_1 and X_2 are independent, then

$$M_{X_1,X_2}(t_1,t_2) = E[e^{t_1X_1+t_2X_2}] = E[e^{t_1X_1}]E[e^{t_2X_2}] = M_{X_1}(t_1)M_{X_2}(t_2).$$

To prove the converse statement: if $M_{X_1,X_2}(t_1,t_2) = M_{X_1}(t_1)M_{X_2}(t_2)$, then

$$\int\int e^{t_1X_1+t_2X_2} f_{X_1,X_2}(x_1,x_2)dx_1dx_2 = \int\int e^{t_1X_1+t_2X_2} f_{X_1}(x_1)f_{X_2}(x_2)dx_1dx_2.$$

By uniqueness of m.g.f., we must have $f_{X_1,X_2}(x_1,x_2) = f_{X_1}(x_1)f_{X_2}(x_2)\ \forall\ x_1,x_2 \in \mathbb{R}$. It follows that X_1 and X_2 are independent.

Theorem 9.5. Let $X_1, X_2, \ldots$ be a sequence of random variables with moment generating function $M_{X_1}, M_{X_2}, \ldots$. If for all t in an open interval containing 0 we have point-wise convergence, i.e.

$$M_{X_n}(t) \to M_X(t),$$

for some random variable X, then X_n converges in distribution to X, i.e.

$$X_n \xrightarrow{D} X.$$

Definition 9.3. Cumulant-Generating function: The cumulants of a random variable X are defined using the cumulant-generating function $K(t)$, which is the natural logarithm of the moment-generating function:

$$K(t) = \ln(M(t)) = \ln(E[e^{tX}]) = \ln(1 + \sum_{i=1}^{\infty} \mu_i' \frac{t^i}{i!}),$$

where $\mu'_i = E[X^i]$, i.e., the ordinary moment of order i (≥ 1). The cumulants k_n, $n \geq 1$, are obtained from a power series expansion of the cumulant generating function:

$$K(t) = \sum_{1}^{\infty} k_n \frac{t^n}{n!} = k_1 \frac{t}{1!} + k_2 \frac{t^2}{2!} + ... = \mu t + \sigma^2 \frac{t^2}{2} + ...$$

This expansion is a Maclaurin's series. Thus, the n-th cumulant can be obtained by differentiating the above expansion n times and evaluating the result at $t = 0$. Therefore, $k_n = K^{(n)}(0)$, where $K^{(n)}(t)$ is the $n$the derivative of $K(t)$ with respect to t. If the moment-generating function does exist, the cumulants can be defined in terms of the relationship between cumulants and momments .

Some basic properties: The n-th cumulant $k_n(X)$ of a random variable X enjoys the following properties:

• if $n > 1$ and c is constant (i.e.not random) then $k_n(X+c) = k_n(X)$, i.e the cumulant is translation invariant.(if $n = 1$ then we have $k_1(X + c) = k_1(X) + c$.)

• if c is a constant (i.e. not random) then $k_n(cX) = C^n k_n(X)$, i.e., the n-th cumulant is homogeneous of degree n.

• If random variables X_1,...,X_m are independent then $k_n(X_1 + ...X_m) = k_n(X_1) + ... + k_n(X_m)$. The moment generating function is given by:

$$M(t) = 1 + \sum_{n=1}^{\infty} \mu'_n \frac{t^n}{n!} = e^{(\sum_{n=1}^{\infty} k_n \frac{t^n}{n!})} = e^{K(t)}.$$

Thus, the cumulant generating function is the logarithm of the moment generating function and is given by

$$K(t) = \sum_{n=1}^{\infty} k_n \frac{t^n}{n!} = \ln(1 + \sum_{n=1}^{\infty} \mu'_n \frac{t^n}{n!}) = \ln(M(t)).$$

The moments can be recovered in terms of the cumulants by evaluating the nth derivative of the $e^{K(t)}$ at $t = 0$. The first few expressions are:

$\mu'_1 = k_1$,
$\mu'_2 = k_2 + k_1^2$,
$\mu'_3 = k_3 + 3k_2k_1 + k_1^3$,
$\mu'_4 = k_4 + 4k_3k_1 + 3k_2^2 + 6k_2k_1^2 + k_1^4$.

Similarly, the n-th cumulant k_n is an n-th degree polynomial in the first n non-central moments. The first few expressions are:

$k_1 = \mu'_1$,
$k_2 = \mu'_2 - {\mu'_1}^2$,
$k_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2{\mu'_1}^3$,
$k_4 = \mu'_4 - 4\mu'_3\mu'_1 - 3{\mu'_2}^2 + 12\mu'_2{\mu'_1}^2 - 6{\mu'_1}^4$.

The "prime" distinguishes the moments μ'_n from the central moments $\mu_n = E[(X - \mu'_1)^n]$. To express the central moments as a functions of the cumulants, just drop from these polynomials as follows:

$\mu 1 = 0$,
$\mu_2 = k_2$,

$\mu_3 = k_3,$
$\mu_4 = k_4 + 3k_2^2.$

10 Inequalities

Theorem 10.1 (**Holder's Inequality**). Let $p > 1, q > 1$ such that $1/p + 1/q = 1$. Then

$$\mathbb{E}[|XY|] \leq (\mathbb{E}[|X|^p])^{\frac{1}{p}}(\mathbb{E}[|Y|^q])^{\frac{1}{q}}.$$

Proof: For $x > 0, log(x)$ is concave. This implies for $x_1, x_2 > 0$,

$$log(tx_1 + (1-t)x_2) \geq tlog(x_1) + (1-t)log(x_2),$$
$$x_1^t x_2^{1-t} \leq tx_1 + (1-t)x_2.$$

Putting $x_1 = \frac{|X|^p}{\mathbb{E}[|X|^p]}, x_2 = \frac{|Y|^q}{\mathbb{E}[|Y|^q]}$, and taking $t = 1/p$, we get

$$|XY| \leq \frac{|X|^p \mathbb{E}[|X|^p]^{1/p-1}\mathbb{E}[|Y|^q]^{1/q}}{p} + \frac{|Y|^q \mathbb{E}[|Y|^q]^{1/q-1}\mathbb{E}[|X|^p]^{1/p}}{q}.$$

Taking the expectation, we get

$$\mathbb{E}[|XY|] \leq (\mathbb{E}[|X|^p])^{\frac{1}{p}}(\mathbb{E}[|Y|^q])^{\frac{1}{q}}.$$

Theorem 10.2 (**Minkowski Inequality**). For $p \geq 1$,

$$\mathbb{E}[|X+Y|^p]^{\frac{1}{p}} \leq \mathbb{E}[|X|^p]^{\frac{1}{p}} + \mathbb{E}[|Y|^p]^{\frac{1}{p}}.$$

Proof: For $p \geq 1$,

$$|X+Y|^p \leq |X||X+Y|^{p-1} + |Y||X+Y|^{p-1}.$$

Taking expectation, and using Holder's inequality with Y replaced by $|X+Y|^{p-1}$ we get

$$\begin{aligned}\mathbb{E}[|X+Y|^p] &\leq \mathbb{E}[|X|^p]^{1/p}\mathbb{E}[|X+Y|^{(p-1)q}]^{1/q} + \mathbb{E}[|Y|^p]^{1/p}\mathbb{E}[|X+Y|^{(p-1)q}]^{1/q} \\ &= (\mathbb{E}[|X|^p]^{1/p} + \mathbb{E}[|Y|^p]^{1/p}) \cdot \mathbb{E}[|X+Y|^{(p-1)q}]^{1/q}\end{aligned}$$

Excluding the trivial case, when $\mathbb{E}[|X+Y|^p] = 0$ and noting that $(p-1)q = p$, dividing by $\mathbb{E}[|X+Y|^p]^{1/q}$, we get

$$\mathbb{E}[|X+Y|^p]^{\frac{1}{p}} \leq \mathbb{E}[|X|^p]^{\frac{1}{p}} + \mathbb{E}[|Y|^p]^{\frac{1}{p}}.$$

Theorem 10.3 (**Cauchy-Schwartz's Inequality**). If X and Y are two random variables, then

$$\mathbb{E}(XY)^2 \leq \mathbb{E}(X^2)\mathbb{E}(Y^2).$$

The equality holds if and only if $X = aY$ for some constant $a \in \mathbb{R}$.
Proof: Clearly, $\mathbb{E}[(tX+Y)^2] \geq 0$ for $t \in \mathbb{R}$. Define a non-negative function from $\mathbb{R}$ to itself by $f(t) = \mathbb{E}[(tX+Y)^2]$. Then

$$f(t) = \mathbb{E}[(tX+Y)^2] = (\mathbb{E}[X^2])t^2 + 2t\mathbb{E}[XY] + \mathbb{E}[Y^2]$$

is a non-negative quadratic function, which implies $\mathbb{E}[X^2] \geq 0$ and $(2\mathbb{E}[XY])^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0$. That is, $\mathbb{E}(XY)^2 \leq \mathbb{E}(X^2)\mathbb{E}(Y^2)$.

The equality holds iff $(2\mathbb{E}[XY]^2)^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] = 0$ iff $f\left(\frac{-\mathbb{E}[XY]}{\mathbb{E}[X^2]}\right) = 0$ iff $\mathbb{E}\left[\left(-X\frac{\mathbb{E}[XY]}{\mathbb{E}[X^2]} + Y\right)^2\right] = 0$ iff $X = \frac{\mathbb{E}[X^2]}{\mathbb{E}[XY]}Y$.

Theorem 10.4. Let $h(X)$ be a non-negative measurable function of a random variable X. If $\mathbb{E}[h(X)]$ exists, then for every $\epsilon > 0$,

$$\mathbb{P}(h(X) \geq \epsilon) \leq \frac{\mathbb{E}[h(X)]}{\epsilon}.$$

Proof: When X is a discrete random variable, let $\mathbb{P}(X = x_k) = p_k,\ k = 1, 2, \ldots,$ then

$$\begin{aligned}\mathbb{E}[h(X)] &= \sum_k h(x_k)p_k = \sum_{k:h(x_k)\geq\epsilon} h(x_k)p_k + \sum_{k:h(x_k)<\epsilon} h(x_k)p_k \\ &\geq \sum_{k:h(x_k)\geq\epsilon} h(x_k)p_k \geq \epsilon \sum_{k:h(x_k)\geq\epsilon} p_k = \epsilon\mathbb{P}(h(X) \geq \epsilon).\end{aligned}$$

Corollary: (**Markov's Inequality:**) Let $h(X) = |X|^r$ and $\epsilon = a^r$, where $r, a > 0$. Then

$$\mathbb{P}(|X| \geq a) \leq \frac{\mathbb{E}[|X|^r]}{a^r}.$$

Corollary: (**Chebyshev's Inequality:**) Let $h(X) = (X - \mu)^2$ and $\epsilon = k^2\sigma^2$, where $\mu = \mathbb{E}[X], \sigma^2 = Var(X)$. Then

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

If X is a random variable with finite mean μ and variance σ^2, then for any value $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Theorem 10.5 (**Jensen's Inequality**)**.** Let $g : \mathbb{R} \to \mathbb{R}$ be a convex function, and X be a random variable, then

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]).$$

Proof: A function g is convex if, for any point x_0

$$g(x) \geq g(x_0) + b(x - x_0),\ \forall\ x$$

where b is the slope of the tangent at point x_0. Setting $x = X$ and $x_0 = \mathbb{E}[X]$, the inequality becomes

$$g(X) \geq g(\mathbb{E}[X]) + b(X - \mathbb{E}[X]).$$

By taking the expectation on both sides, we get

$$\mathbb{E}[g(X)] \geq \mathbb{E}[g(\mathbb{E}[X]) + b(X - \mathbb{E}[X])] = g(\mathbb{E}[X]) + b(\mathbb{E}[X] - \mathbb{E}[X]) = g(\mathbb{E}[X]).$$

Example 10.1. If X has mean μ and standard deviation σ, the ratio $r = \frac{|\mu|}{\sigma}$ is called the **measurement signal-to-noise ration** of X. The idea being that X can be expressed as $X = \mu + (X - \mu)$ with μ representing the signal and $(X - \mu)$ representing the noise. If we define $D = |\frac{X-\mu}{\mu}|$ as the **relative deviation** of X from its signal or mean, show that, for any $\alpha > 0$,

$$\mathbb{P}(D < \alpha) \geq 1 - \frac{1}{r^2\alpha^2}.$$

Proof: By Chebyshev's Inequality,

$$\mathbb{P}(D < \alpha) = 1 - \mathbb{P}(D \geq \alpha) = 1 - \mathbb{P}(|X - \mu| \geq |\mu|\alpha) \geq 1 - \frac{\sigma^2}{|\mu|^2\alpha^2} = 1 - \frac{1}{r^2\alpha^2}.$$

Example 10.2. Compute the measurement signal-to-noise ratio, i.e., $\frac{|\mu|}{\sigma}$ where $\mu = \mathbb{E}[X]$ and $\sigma^2 = Var(X)$, of the following random variables:

(a) $Poisson(\lambda)$

(b) $Bin(n, p)$

(c) $Geometric(p)$

(d) $exp(\lambda)$

(e) $N(\mu, \sigma^2)$.

11 Law of Large Numbers and Central Limit Theorem

Let $X_1, X_2, \ldots$ be an infinite sequence (population) of *i.i.d.* random variables with $\mathbb{E}[X_i] = \mu$. Then both versions of the law of large version states that the sample mean of the population,

$$\overline{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

converges to the expected value μ. That is,

$$\overline{X}_n \longrightarrow \mu.$$

An assumption of finite variance is not necessary. Large or infinite variance will make the convergence slower, but the laws holds anyway. Note that, for $Var(X_i) = \sigma^2$

$$\mathbb{E}[\overline{X}_n] = \mathbb{E}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right] = \frac{\sum_{i=1}^{n} \mathbb{E}[X_i]}{n} = \frac{n\mu}{n} = \mu,$$

$$Var(\overline{X}_n) = Var\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{\sum_{i=1}^{n} Var(X_i)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Theorem 11.1 (**Strong Law of Large Numbers:**). Let $X_1, X_2, \ldots$ be an infinte sequence of *i.i.d.* random variables with $\mathbb{E}[X_i] = \mu \ \forall \ i$. Then the sample mean converges almost surely to the mean μ, i.e.,

$$\mathbb{P}\Big(\lim_{n\to\infty} \overline{X}_n = \mu\Big) = 1.$$

Proof: Probability and Random Process, Third Edition, Grimmett and Satirzaker. Section-7.5- The strong law, Page 329.

Theorem 11.2 (**Weak Law of Large Numbers:**). Let $X_1, X_2, \ldots$ be an infinite sequence of *i.i.d.* random variables with $\mathbb{E}[X_i] = \mu \ \forall \ i$. Then the sample mean converges in probability to the mean μ, i.e.,

$$\lim_{n\to\infty} \mathbb{P}\Big(|\overline{X}_n - \mu| \geq \epsilon\Big) = 0.$$

Proof: For finite variance case: Using Chebyshev's inequality, for given $\epsilon > 0$,

$$\mathbb{P}\Big(|\overline{X}_n - \mu| \geq \epsilon\Big) \leq \frac{Var(\overline{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon}.$$

As $n \to \infty$, it follows that

$$\lim_{n\to\infty} \mathbb{P}\Big(|\overline{X}_n - \mu| \geq \epsilon\Big) = 0.$$

For finite or infinite variance case: Since X_i are iid random variables, X_i/n are also iid random variables for $i = 1, 2 \ldots, n$. The characteristic function of $\overline{X}_n$ is given by

$$\begin{aligned}\phi_{\overline{X}_n}(t) &= \phi_{\frac{1}{n}\sum_{i=1}^n X_i}(t) = \phi_{\sum_{i=1}^n \frac{X_i}{n}}(t) = (\phi_{X/n}(t))^n = (\phi_X(t/n))^n \\ &= (\mathbb{E}[e^{\frac{\iota t X}{n}}])^n = \left(1 + \frac{\iota t}{n}\mathbb{E}[Y] + o\Big(\frac{t}{n}\Big)\right)^n\end{aligned}$$

Thus,

$$\lim_{n\to\infty} \phi_{\overline{X}_n}(t) = \lim_{n\to\infty} \left(1 + \frac{\iota t\mu}{n} + o\left(\frac{t}{n}\right)\right)^n = e^{\iota t\mu} = \mathbb{E}[e^{\iota t\mu}] = \phi_\mu(t).$$

Since, $\phi_{\overline{X}_n}(t)$ converges point-wise to $\phi_\mu(t)$, by Levy continuity theorem, $\overline{X}_n$ converges in distribution to μ. Because μ is a fixed constant, the convergence in distribution imply convergence in probability.
For more details about characteristics and Levy continuity theorem, see Appendix.

Almost sure convergence is also called strong convergence of random variables. The strong law implies the weak law but not the vice-versa. Interpreting this result, the weak law essentially states that for any non-zero margin specified, no matter how small. With a sufficiently large sample, there will be a very high probability that the average of the observations will be close to the expected value, that is, within the margin convergence in probability; also called the **weak convergence of random variables.** This version is called the weak law because random variables may converge weakly (in probability) as above without converging strongly (in almost surely).

Theorem 11.3 (**Central Limit Theorem:**). If $X_1, X_2, \ldots, X_n$ constitute a random sample from an infinite population with mean μ and finite variance $\sigma^2 < \infty$, then

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{D} N(0,1) \text{ as } n \to \infty.$$

Proof: Let $Z = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} = \frac{X_1-\mu}{\sqrt{n}\sigma} + \frac{X_2-\mu}{\sqrt{n}\sigma} + \cdots + \frac{X_n-\mu}{\sqrt{n}\sigma}$. Since $X_i, i = 1, 2, \ldots, n$, are iid random variables, $\frac{X_i-\mu}{\sqrt{n}\sigma}, i = 1, 2, \ldots, n$, are iid random variables. The moment generating function of Z is given by

$$M_Z(t) = M_{\sum_{i=1}^n \frac{X_i-\mu}{\sqrt{n}\sigma}}(t) = \left(M_{\frac{X-\mu}{\sqrt{n}\sigma}}(t)\right)^n = \left(M_{(X-\mu)}\Big(\frac{t}{\sqrt{n}\sigma}\Big)\right)^n.$$

Expanding the MGF $M_{(X-\mu)}\left(\frac{t}{\sqrt{n}\sigma}\right)$ about $t=0$ gives,

$$\begin{aligned}
M_{(X-\mu)}\left(\frac{t}{\sqrt{n}\sigma}\right) &= \mathbb{E}\left[e^{\frac{t(X-\mu)}{\sqrt{n}\sigma}}\right] = \left(M_{(X-\mu)}\left(\frac{t}{\sqrt{n}\sigma}=0\right)\right) + \left(\frac{M'_{(X-\mu)}\left(\frac{t}{\sqrt{n}\sigma}=0\right)}{1!}\right)\left(\frac{t}{\sqrt{n}\sigma}\right) \\
&\quad + \left(\frac{M''_{(X-\mu)}\left(\frac{t}{\sqrt{n}\sigma}=0\right)}{2!}\right)\left(\frac{t}{\sqrt{n}\sigma}\right)^2 + \left(\frac{M'''_{(X-\mu)}\left(\frac{t}{\sqrt{n}\sigma}=0\right)}{3!}\right)\left(\frac{t}{\sqrt{n}\sigma}\right)^3 + \dots \\
&= 1 + \left(\frac{t}{\sqrt{n}\sigma}\right)(\mathbb{E}[X-\mu]) + \left(\frac{t^2}{2n\sigma^2}\right)(\mathbb{E}[(X-\mu)^2]) + \left(\frac{t^3}{6n^{3/2}\sigma^3}\right)(\mathbb{E}[(X-\mu)^3]) + \cdots \\
&= 1 + \left(\frac{t}{\sqrt{n}\sigma}\right)(\mathbb{E}[X]-\mu) + \left(\frac{t^2}{2n\sigma^2}\right)(\mathbb{E}[(X-\mu)^2]) + O(t^3) \\
&\approx 1 + \left(\frac{t}{\sqrt{n}\sigma}\right)(\mu-\mu) + \left(\frac{t^2}{2n\sigma^2}\right)(\sigma^2) \\
&= 1 + \frac{t^2}{2n}.
\end{aligned}$$

Using the last two calculation and the fact that $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e^x$, we get

$$\lim_{n\to\infty} M_Z(t) = \lim_{n\to\infty}\left(1+\frac{t^2}{2n}\right)^n = e^{\frac{t^2}{2}} = M_{N(0,1)}(t).$$

$M_Z(t)$ converges point-wise to $M_Z(t)$, implies Z converges in distribution to $N(0,1)$; that is,

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{D} N(0,1) \text{ as } n\to\infty.$$

Exercise:

(a) If X is $bin(n,p)$, then show that $\frac{X-\mathbb{E}[X]}{\sqrt{Var(X)}}$ is standard normal as $n\to\infty$, where neither p nor $q=1-p$ is very small

(b) If X is either Poisson with parameter λ or $\Gamma(1,\lambda)$, show that the distribution of $Y_\lambda = \frac{X-\mathbb{E}(X)}{Var(X)}$ approaches the $N(0,1)$ distribution as $\lambda\to\infty$

(c) Show that i) if $X_n \overset{D}{\sim} Bin(n,\lambda/n)$, then the distribution of X_n converges to a Poisson distribution when $n\to\infty$,
ii) if $Y_n \overset{D}{\sim} Geometric(\lambda/n)$, then the distribution of $\frac{Y_n}{n}$ converges to an exponential distribution.

12 Chi-square Distribution

Definition 12.1 (Chi-square distribution)**.** A random variable X has the **chi-square distribution** with v degrees of freedom if its pdf is given by

$$f(x) = \begin{cases} \frac{1}{2^{v/2}} x^{\frac{v-2}{2}} e^{-\frac{1}{2}}, & for\ x > 0 \\ 0, & elsewhere. \end{cases}$$

We write a chi-square distribution with v degree of freedom as $\chi^2(v)$.

$M_X(t) = (1-2t)^{-\frac{1}{2}}$.
$\mathbb{E}[X] = M'(0) = -\dfrac{v}{2}(1-2t)^{-\frac{v}{2}-1}(-2)\big|_{t=0} = v$.
If $X \overset{D}{\sim} N(0,1)$, then $X^2 \overset{D}{\sim} \chi^2(1)$.

Theorem 12.1. If $X_1, X_2, \ldots, X_n$ are independent random variables having $N(0,1)$ distribution, then $Y = \sum_{i=1}^n X_i^2$ has the χ^2 distribution with $v = n$ degree of freedom.
Proof: The moment generating function of Y can be written as:

$$M_Y(t) = M_{\sum_{i=1}^n X_i^2}(t) = [M_{X_i^2}(t)]^n = [M_{\chi^2(1)}]^n = (1-2t)^{-\frac{n}{2}} = M_{\chi^2(n)}(t).$$

Hence, $Y = \sum_{i=1}^n X_i^2$ has chi-square distribution with n degrees of freedom.

Similarly, we can obtain; if $X_i \overset{D}{\sim} \chi^(v_i)\ for\ i = 1, 2, \ldots, n$ are independent random variables, then $Y = \sum_{i=1}^n \overset{D}{\sim} \chi^2(v_1 + v_2 + \ldots + v_n)$.

Theorem 12.2. If X_1 and X_2 are independent random variables such that $X_1 \overset{D}{\sim} \chi^2(v_1)$ and $X_1 \overset{D}{\sim} \chi^2(v_1)$ with $v > v_1$, then show that $X_2 \overset{D}{\sim} \chi^2(v - v_1)$.
Proof: Since X_1 and X_2 are independent random variables, $M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t)$,

$$M_{X_2}(t) = \frac{M_{X_1+X_2}(t)}{M_{X_1}(t)} = \frac{(1-2t)^{v/2}}{(1-2t)^{v_1/2}} = (1-2t)^{\frac{v-v_1}{2}}.$$

Thus, $X_2 \overset{D}{\sim} \chi^2(v - v_1)$.

The sample variance of a random sample of size n from a population is defined as

$$S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})^2.$$

Theorem 12.3. If $\overline{X}$ and S^2 are the sample mean and sample variance of a random sample of size n from a normal population with mean μ and variance σ^2, then

(a) $\overline{X}$ and S^2 are independent,

(b) $\frac{(n-1)S^2}{\sigma^2} \overset{D}{\sim} \chi^2(n-1)$.

Proof: (a) Let $X_1, X_2, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ random variables. Then $\overline{X}$ and $(X_1 - \overline{X}), (X_2 - \overline{X}), \ldots, (X_n - \overline{X})$ are independent. We compute the joint joint MGF of $\overline{X}$ and $X_1 - \overline{X}, X_2 - \overline{X}, \ldots, X_n - \overline{X}$ as follows:

$$\begin{aligned}
M(t, t_1, t_2, \ldots, t_n) &= \mathbb{E}[e^{t\overline{X} + t_1(X_1 - \overline{X}) + t_2(X_2 - \overline{X}) + \cdots + t_n(X_n - \overline{X})}] = \mathbb{E}[e^{\sum_{i=1}^n t_i X_i - (\sum_{i=1}^n t_i - t)\overline{X}}] \\
&= \mathbb{E}[e^{\sum_{i=1}^n X_i (t_i - \frac{t_1 + t_2 + \cdots + t_n - t}{n})}] = \mathbb{E}\left[\prod_{i=1}^n e^{X_i \left(\frac{nt_i - n\bar{t} + t}{n}\right)}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{X_i \left(\frac{nt_i - n\bar{t} + t}{n}\right)}\right] \\
&= \prod_{i=1}^n e^{\mu} e^{\mu \frac{(t - n(t_i - \bar{t}))}{n}} + \frac{\sigma^2}{2} \frac{1}{n^2} (t + n(t_i - \bar{t}))^2 = e^{\mu t} \cdot e^{\frac{\sigma^2}{2n^2}(nt^2 + n^2 \sum_{i=1}^n (t_i - \bar{t})^2)} \\
&= e^{\mu t + \frac{\sigma^2}{2n} t^2} \cdot e^{\frac{\sigma^2}{2} \sum_{i=1}^n (t_i - \bar{t})^2} = M_{\overline{X}}(t) M_{X_1 - \overline{X}, X_2 - \overline{X}, \ldots, X_n - \overline{X}}(t_1, t_2, \ldots, t_n) \\
&= M(t, 0, 0, \ldots, 0) M(0, t_1, t_2, \ldots, t_n).
\end{aligned}$$

Hence, $\overline{X}$ & S^2 are independent.
(b) The random variable $\frac{(n-1)S^2}{\sigma^2}$ has a χ^2 distribution with $(n-1)$ degrees of freedom, i.e., $\chi^2(n-1)$.

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \overline{X})^2 + n(\overline{X} - \mu)^2.$$

Divide each term by σ^2 and substitute $(n-1)S^2$ for $\sum_{i=1}^n (X_i - \overline{X})^2$, it follows that

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}\right)^2. \qquad (*)$$

We know

$$\begin{aligned}
X_i &\overset{D}{\sim} N(\mu, \sigma^2) \\
\Rightarrow \frac{X_i - \mu}{\sigma} &\overset{D}{\sim} N(0, 1) \\
\Rightarrow \left(\frac{X_i - \mu}{\sigma}\right)^2 &\overset{D}{\sim} \chi^2(1) \\
\Rightarrow \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 &\overset{D}{\sim} \chi^2(n).
\end{aligned}$$

We also know that,

$$\begin{aligned}
\overline{X} &\overset{D}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \\
\left(\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}\right) &\overset{D}{\sim} N(0, 1) \\
\left(\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}\right) &\overset{D}{\sim} \chi^2(1).
\end{aligned}$$

Therefore, using equation $(*)$, one may conclude that $\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 \overset{D}{\sim} \chi^2(n)$, $\left(\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}\right) \overset{D}{\sim} \chi^2(1)$, and

$$\frac{(n-1)S^2}{\sigma^2} \overset{D}{\sim} \chi^2(n-1).$$

13 Appendix

Characteristic Function:
For a random variable X, its characteristic function is defined as $\phi_X(t) = \mathbb{E}[e^{\iota t X}]$.
$\phi_{X+Y}(t) = \phi_X(t) + \phi_Y(t)$, when X, Y are independent,
$\phi_{X/n}(t) = \mathbb{E}[e^{\iota t X/n}] = \phi_X(t/n)$.

Levy-Continuity Theorem
Suppose we have a sequence of random variables $\{X_n\}_{n=1}^{\infty}$, not necessarily sharing a common probability space, and their corresponding characteristic functions $\{\phi_n\}_{n=1}^{\infty}$. If the $\phi_n(t) \to \phi(t)\ \forall\ t \in \mathbb{R}$ (point-wise convergence), then the following statements hold:

(a) $\phi(t)$ is a characteristic function of some random variable X,

(b) X_n converges in distribution to the random variable X,

(c) $\phi(t)$ is a continuous function of t.

Beta Distribution: $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.
$\Gamma(a+1) = a\Gamma(a)$.
$\int_0^1 x^{\frac{1-\alpha}{2}-1}(1-x)^{\frac{1+\alpha}{2}-1}dx$ converges for $\alpha < 1$ and diverges for $\alpha \geq 1$. Cauchy Distribution.